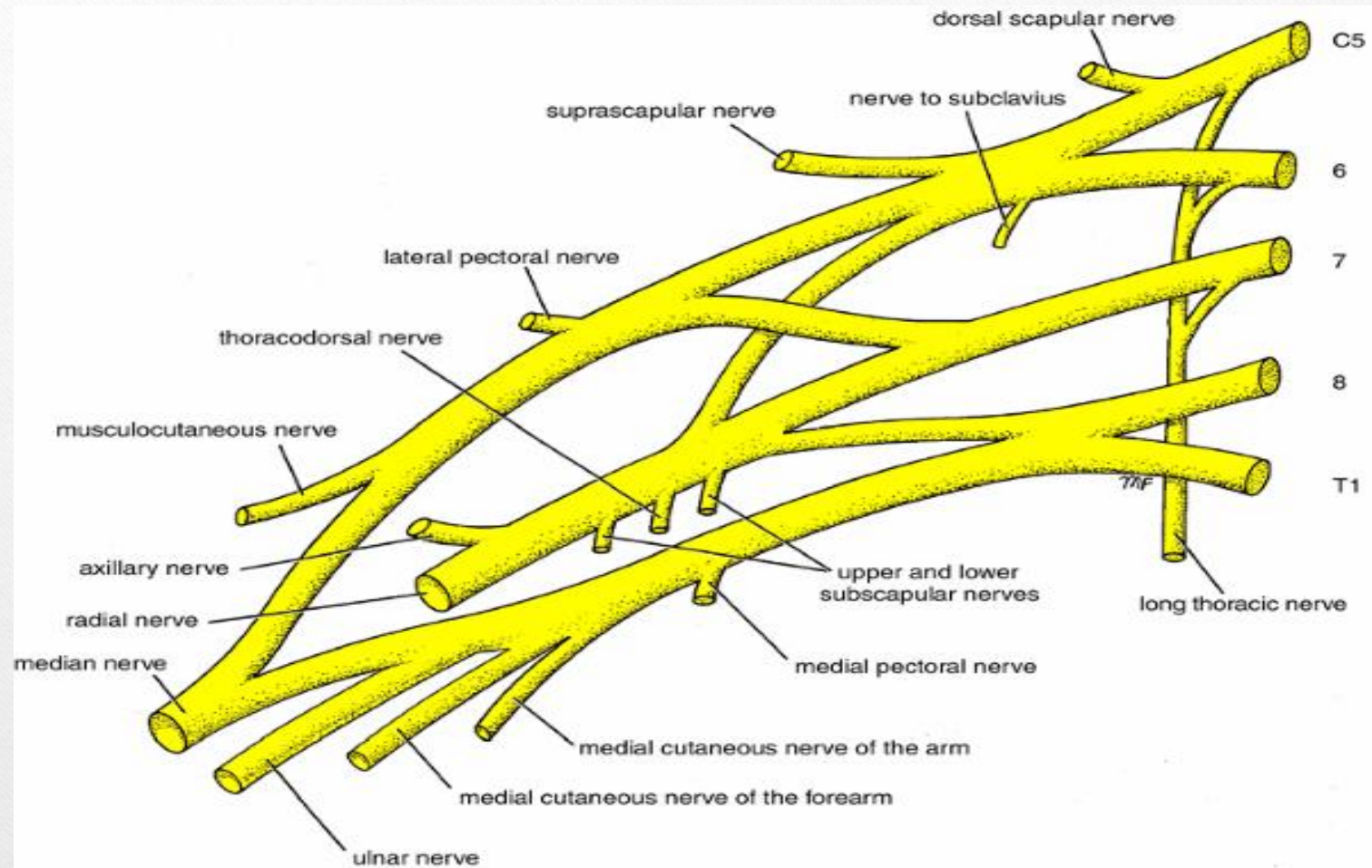
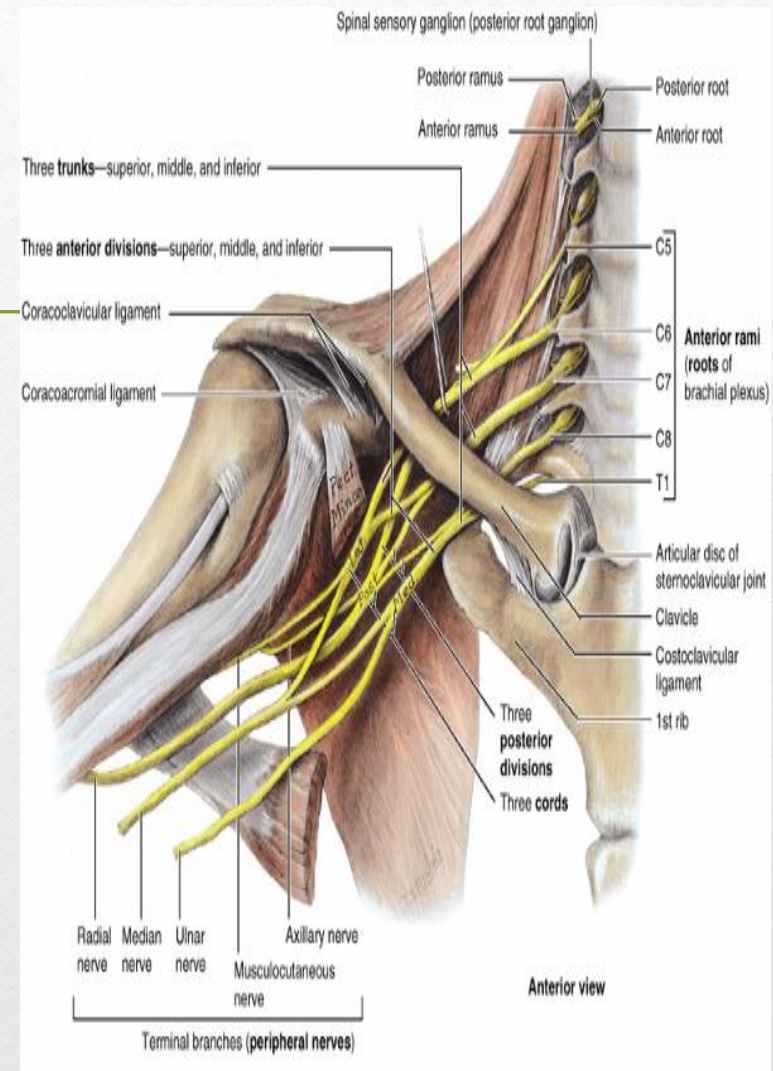


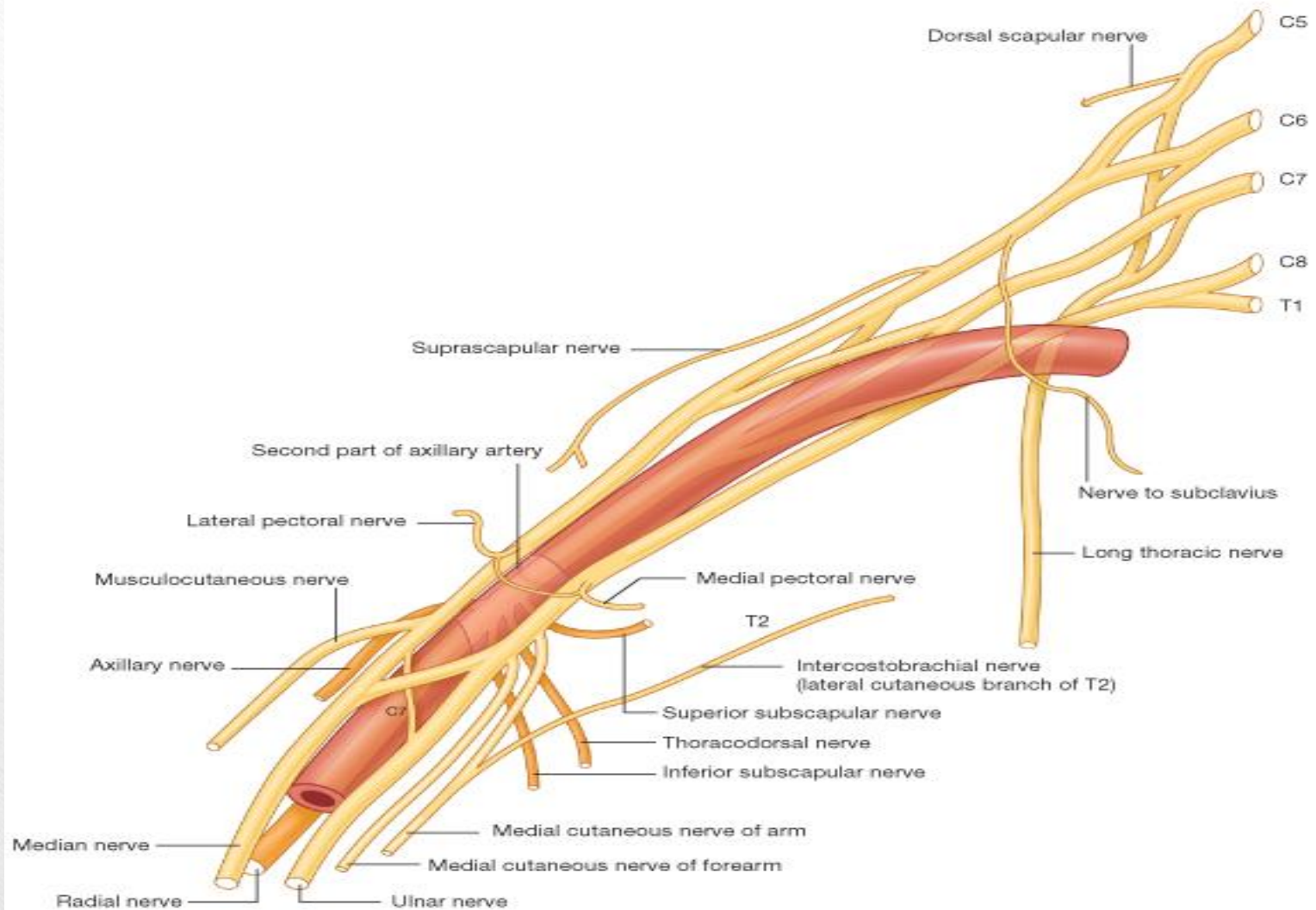
Brachial plexus



Roots, trunks, divisions, cords, and terminal branches of the brachial plexus

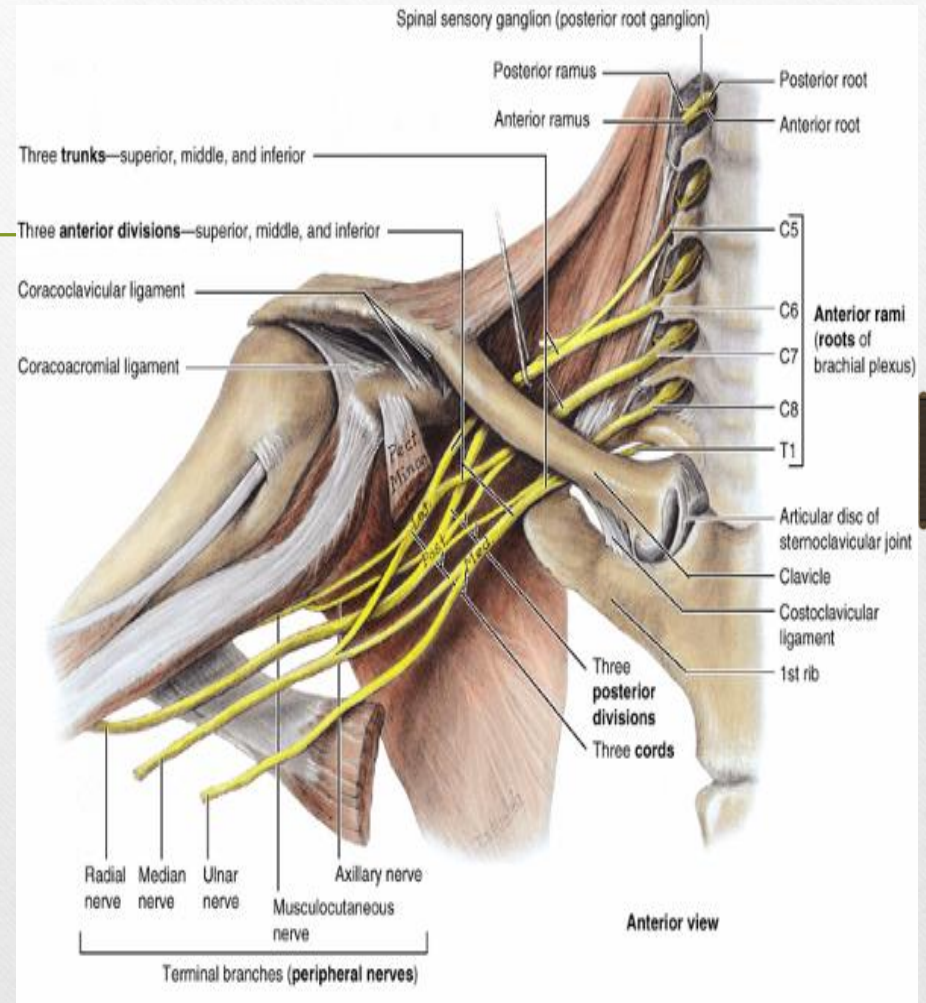
- Each trunk divides into **anterior and posterior**.
- The BP is divided into **supraclavicular** and **infraclavicular** parts by the clavicle
- Four branches of the **supraclavicular** part of the plexus arise from the roots (anterior rami) and trunks of the brachial plexus (**dorsal scapular nerve**, **long thoracic nerve**, nerve to **Subclavius**, and **suprascapular nerve**)





Brachial plexus

- The divisions of the trunks form three cords of the brachial plexus, within the axilla
- Anterior divisions of the superior and middle trunks unite to form the **lateral cord**.
- Anterior division of the inferior trunk continues **as the medial cord**.
- Posterior divisions of all three trunks unite to form the **posterior cord**.



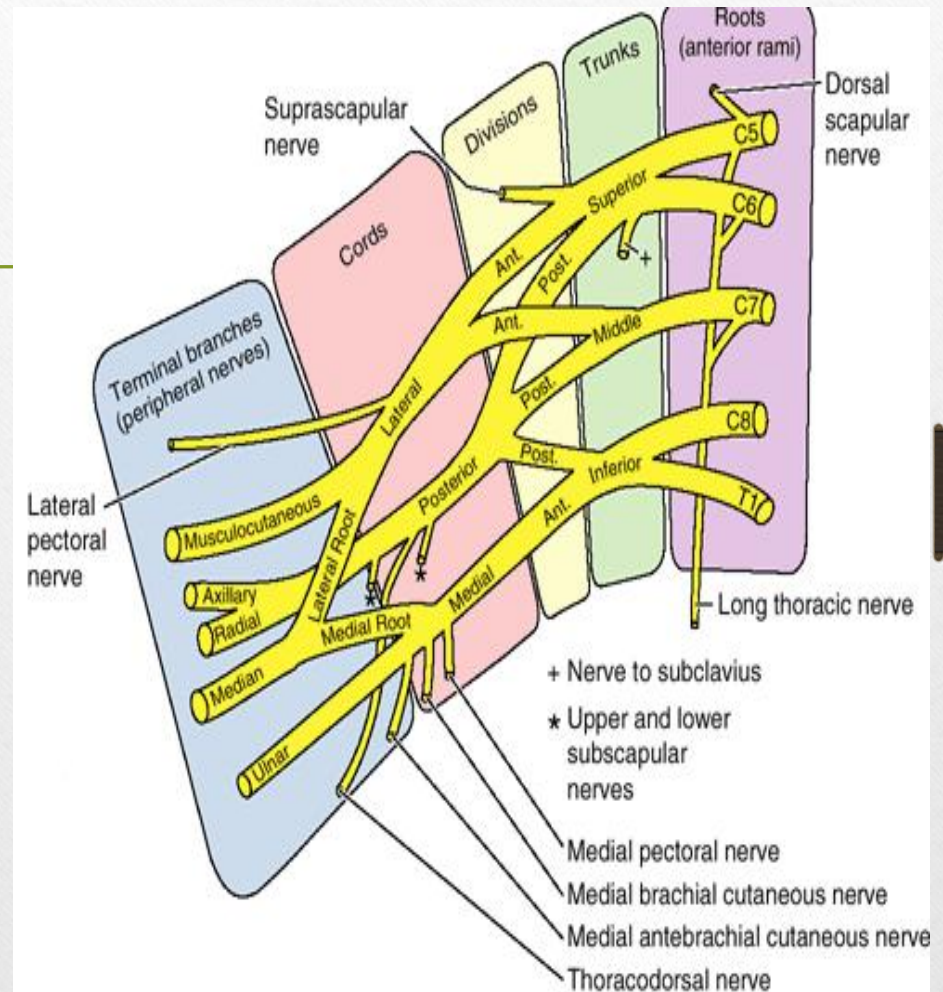
Branches of brachial plexus

Branches of the **lateral cord**.

- Musculocutaneous nerve
- Lateral pectoral nerve
- Lateral root of median nerve

Branches of the **Median cord**

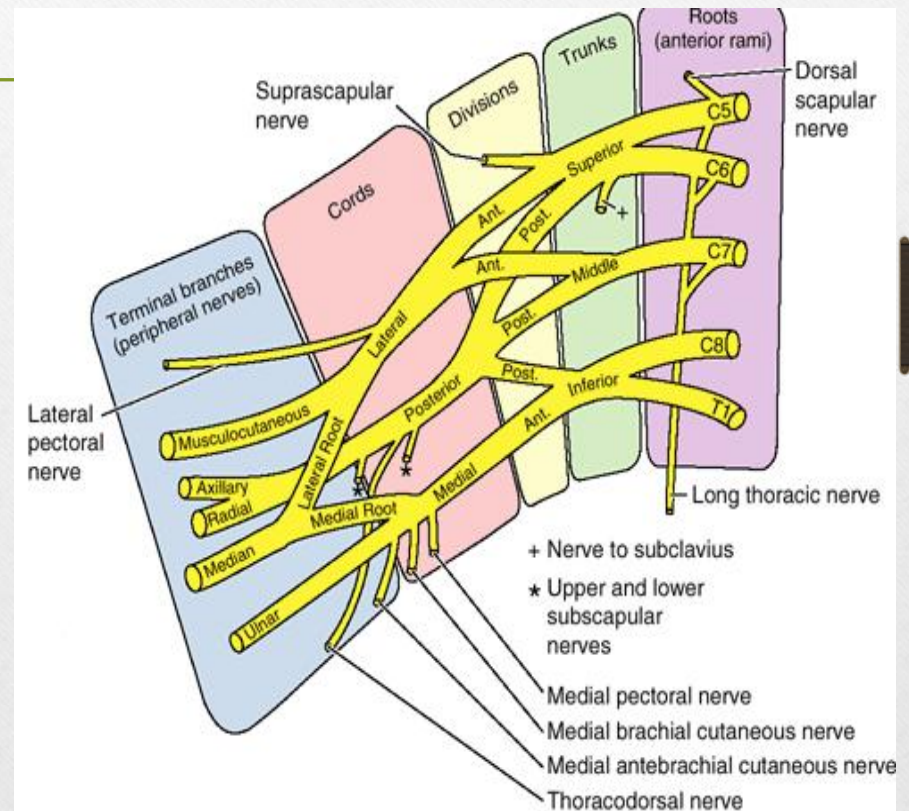
- Medial pectoral nerve
- Medial cutaneous nerve of arm
- Medial cutaneous n.of forearm
- Medial root of Median nerve
- Ulnar nerve



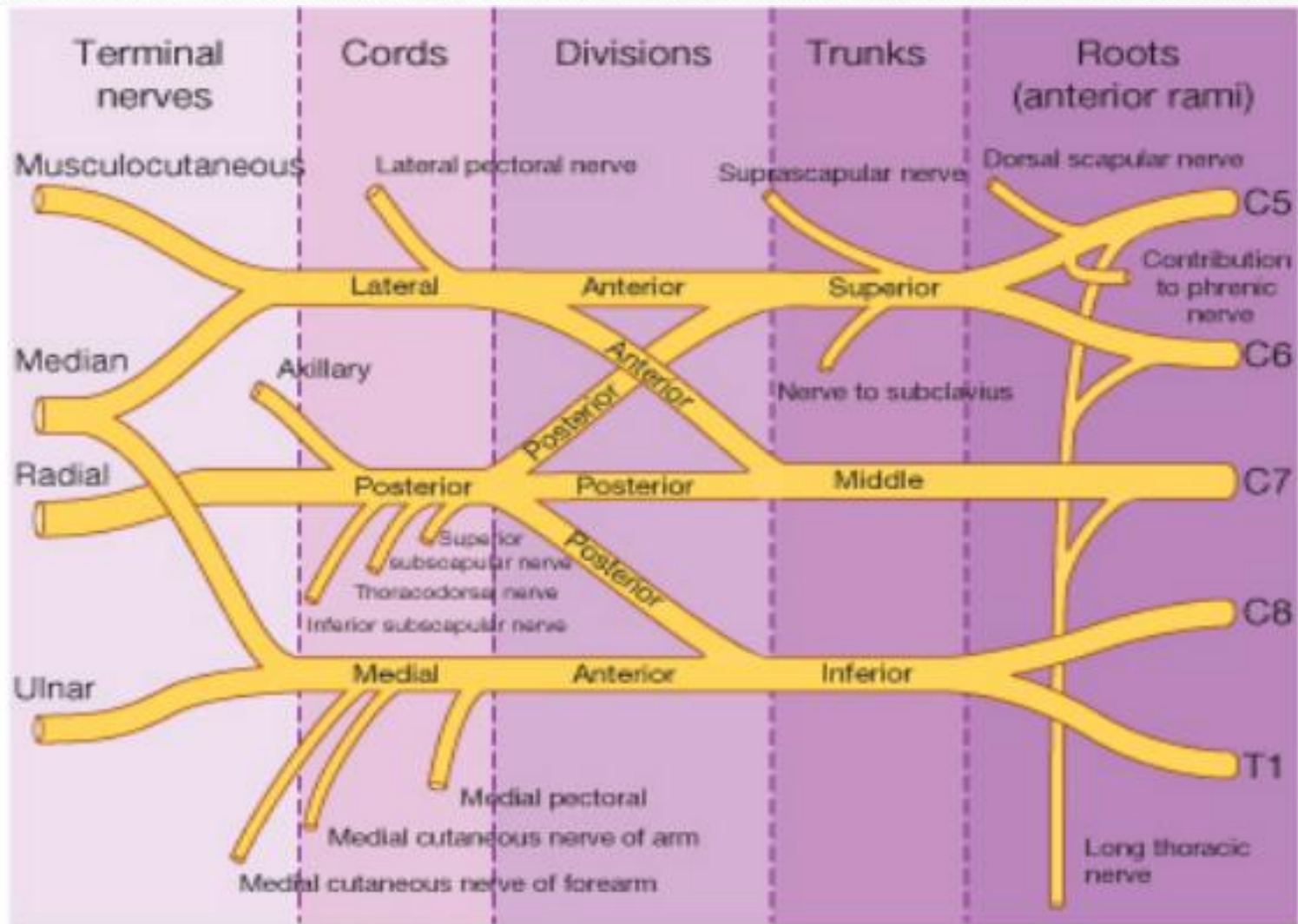
Branches of brachial plexus

➤ Branches of Posterior cord

- Superior subscapular n.
- Thoracodorsal n.
- Inferior subscapular n.
- Axillary n.
- Radial n.

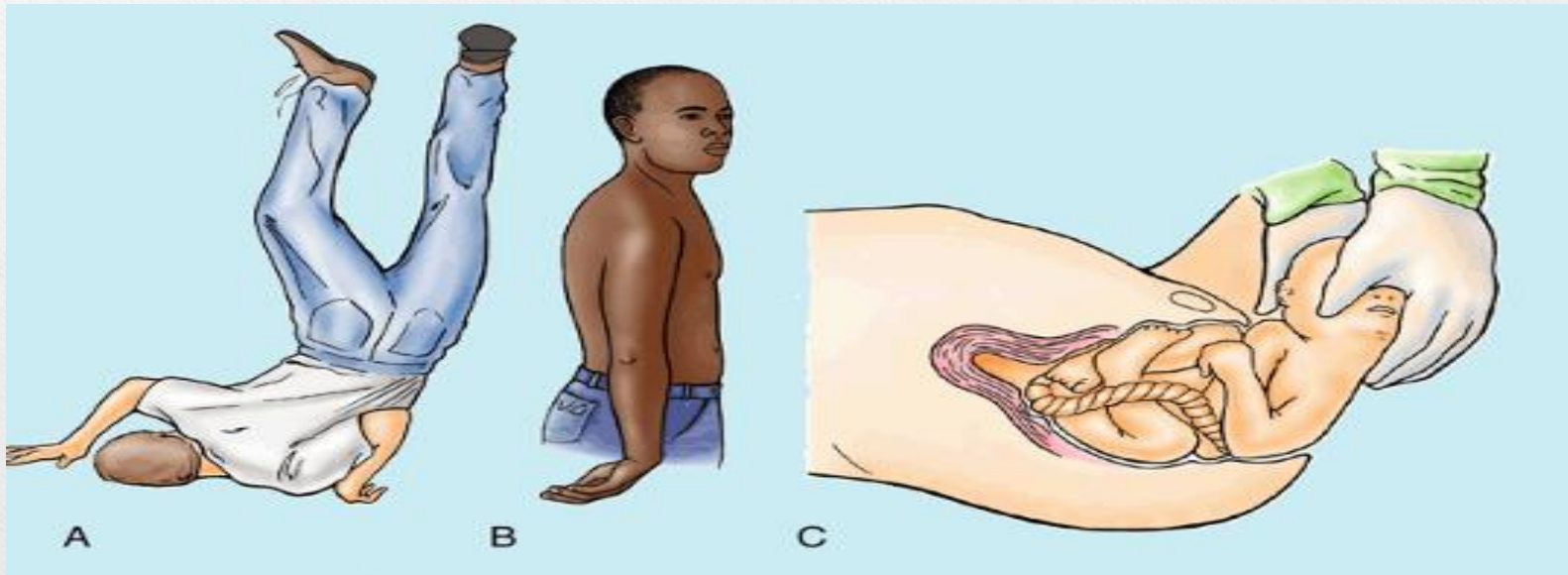


A



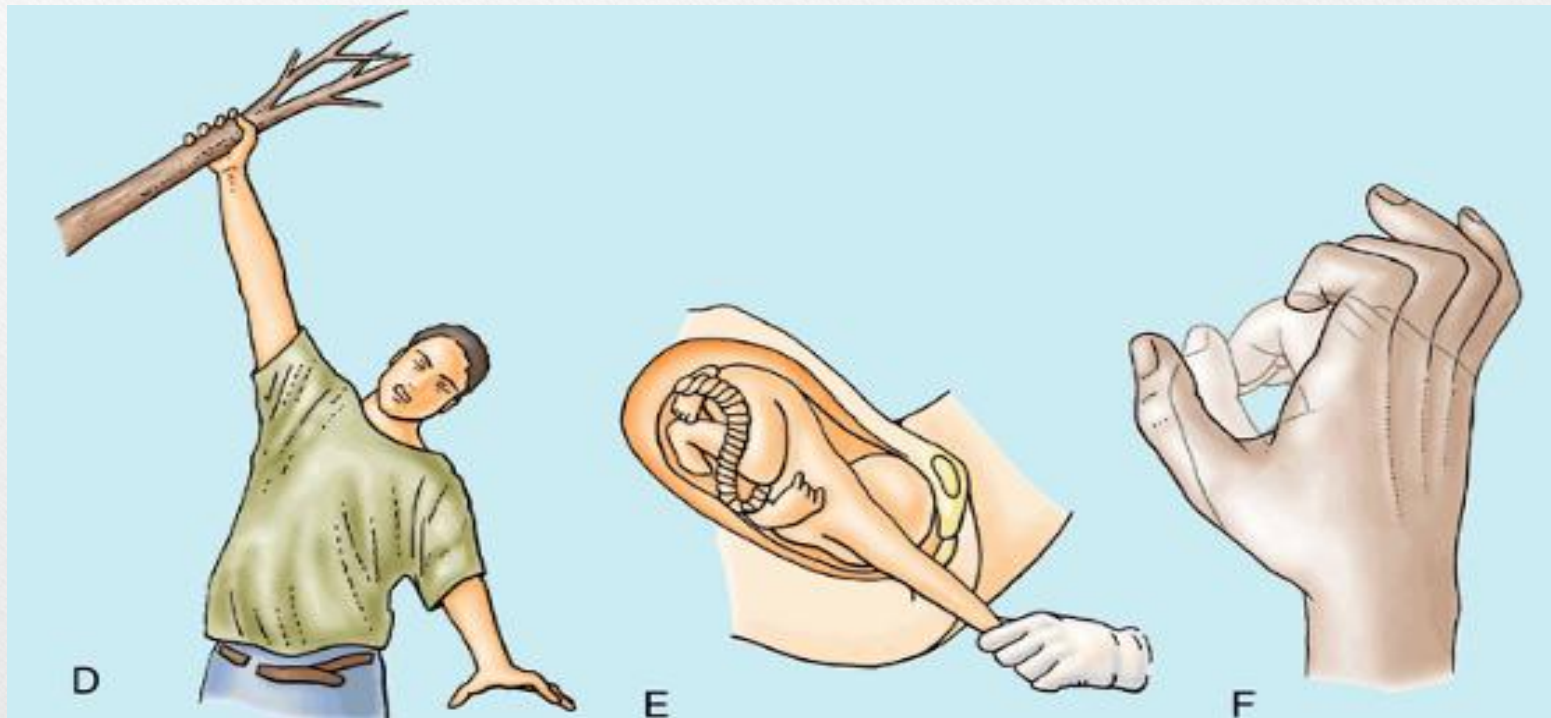
Brachial plexus Injury

- Injuries to superior parts of the brachial plexus (C5 and C6) (**Erb-Duchenne palsy**), usually result from an excessive increase in the angle between the neck and the shoulder.
- characteristic position of the limb **waiter's tip position**
- The usual clinical appearance is an upper limb with **an adducted shoulder, medially rotated arm, and extended elbow.**



Brachial plexus Injury

- Injuries to inferior parts of the brachial plexus (Klumpke paralysis)



Variations of the Brachial Plexus

- **Prefixed brachial plexus:-** When the superiormost root (anterior ramus) of the plexus is C4 and the inferiormost root is C8
- **Postfixed brachial plexus:-** when the superior root is C6 and the inferior root is T2

Arm

The muscles are divided into anterior (flexor) and posterior (extensor) groups by intermuscular septum

✓ **Three flexors**

✓ biceps brachii, brachialis, and coracobrachialis

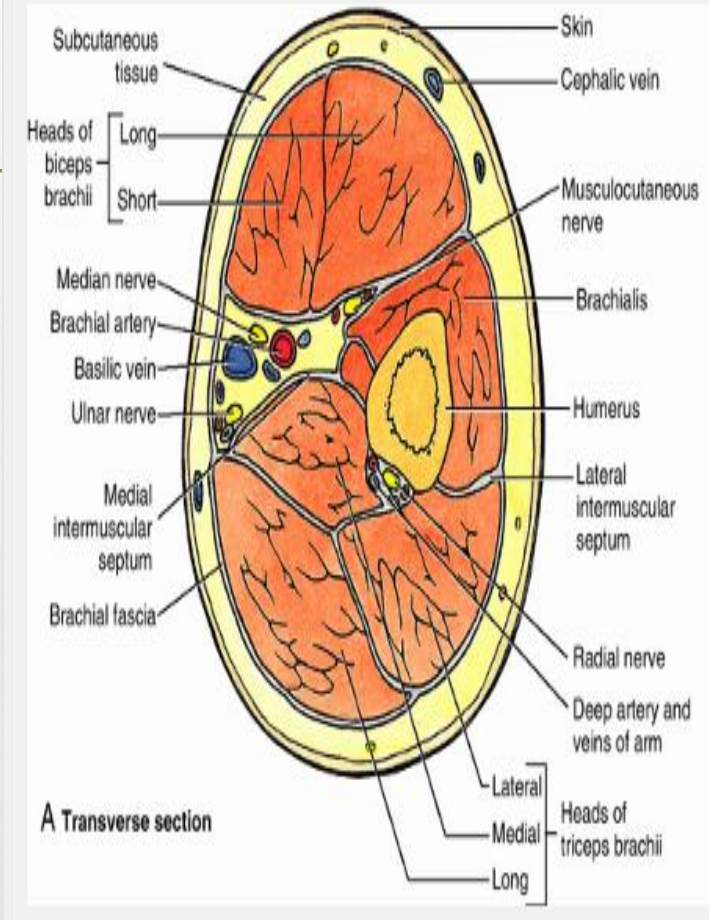
✓ supplied by musculocutaneous nerve

✓ **One extensor**

✓ triceps brachii

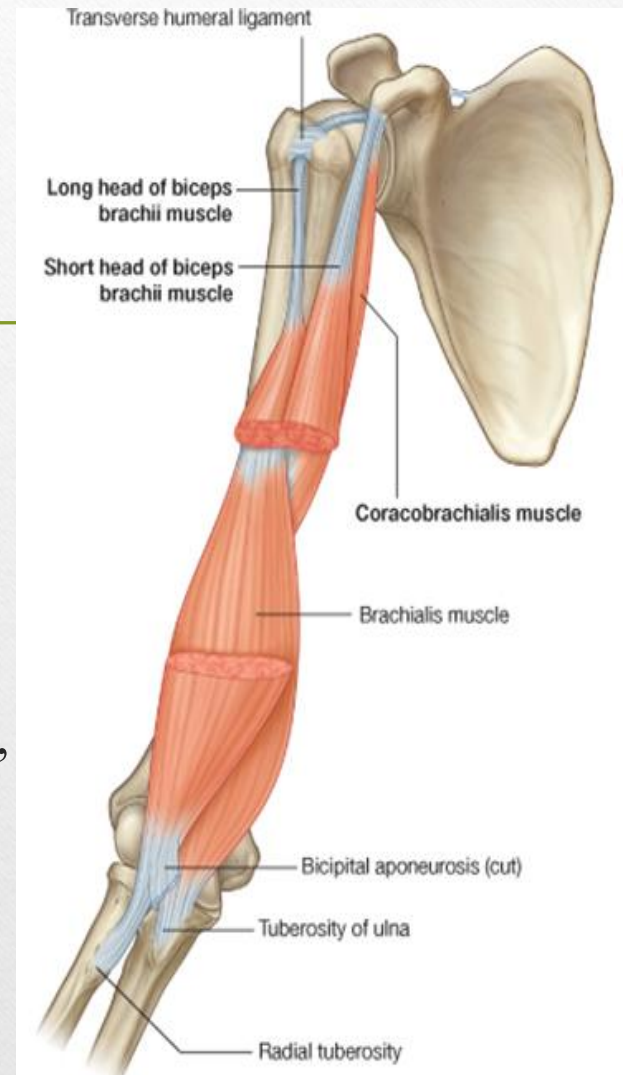
✓ supplied by radial nerve

✓ **Anconeus**



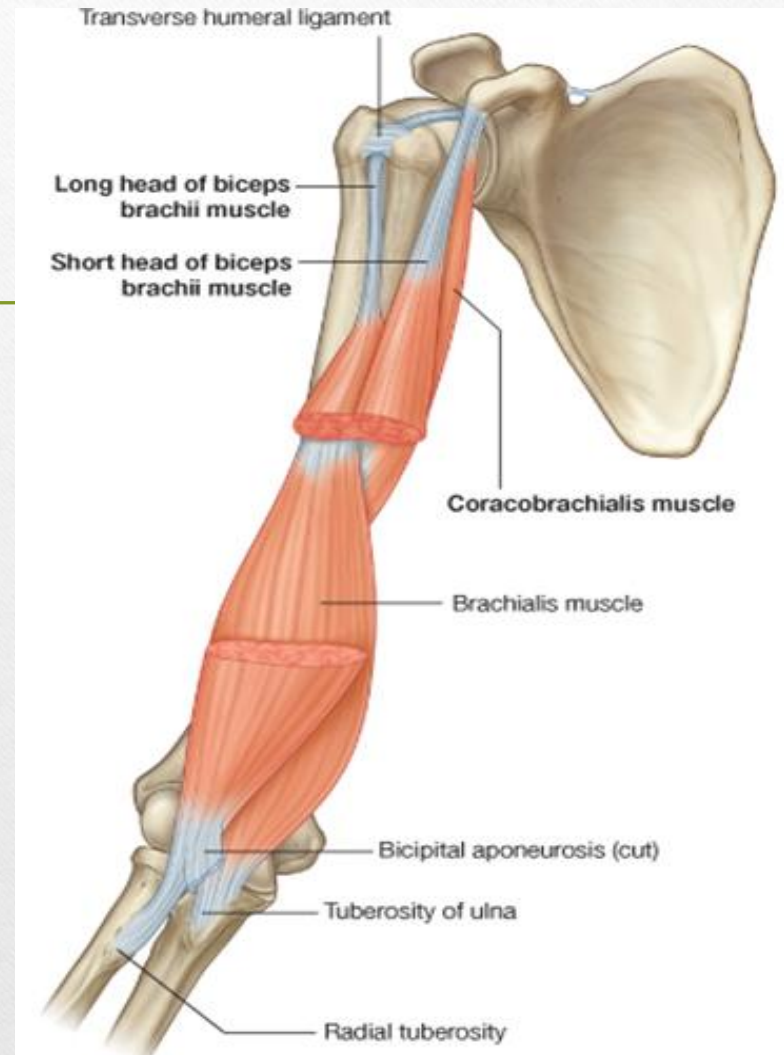
Biceps brachii

- Has two heads; long head and short head
- **Origin:**
 - Short head: tip of coracoid process of scapula
 - Long head: supraglenoid tubercle of scapula
- **Insertion:** tuberosity of radius and fascia of forearm via bicipital aponeurosis
- **Innervation:** Musculocutaneous nerve (C5, C6)
- **Action:** Supinates forearm and, and when it is supine, flexes forearm; short head resists dislocation of shoulder



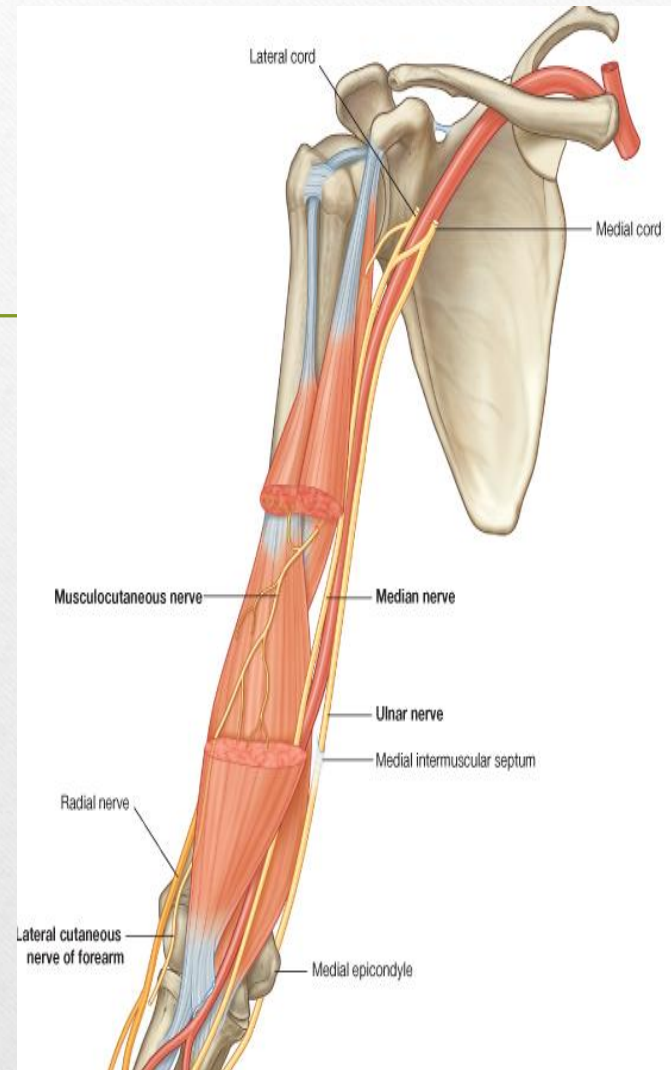
Brachialis

- Lies posterior (deep) to the biceps
- It is the only **pure flexor**
- **Origin:** Distal half of anterior surface of humerus
- **Insertion:** Coronoid process and tuberosity of ulna
- **Innervation:** Musculocutaneous nerve (C5, C6)
- **Action:** Flexes forearm in all positions
- It is the workhorse among the flexor mm of the elbow joint.



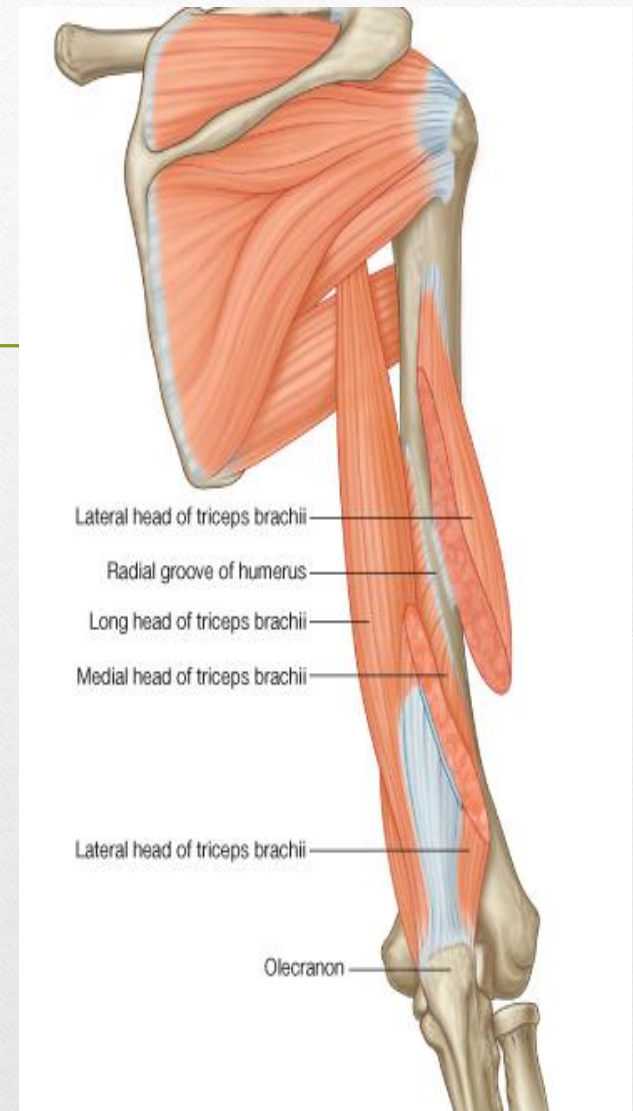
Coracobrachialis

- Elongated muscle in the superomedial part of the arm
 - Useful landmark for locating other structures in the arm
 - musculocutaneous nerve pierces it
 - distal part of its attachment indicates the location of the nutrient foramen of the humerus
- **Origin:** Tip of coracoid process of scapula
- **Insertion:** Middle third of medial surface of humerus
- **Innervation:** Musculocutaneous nerve
- **Action:** Helps flex and adduct arm at the shoulder joint; stabilize shoulder joint



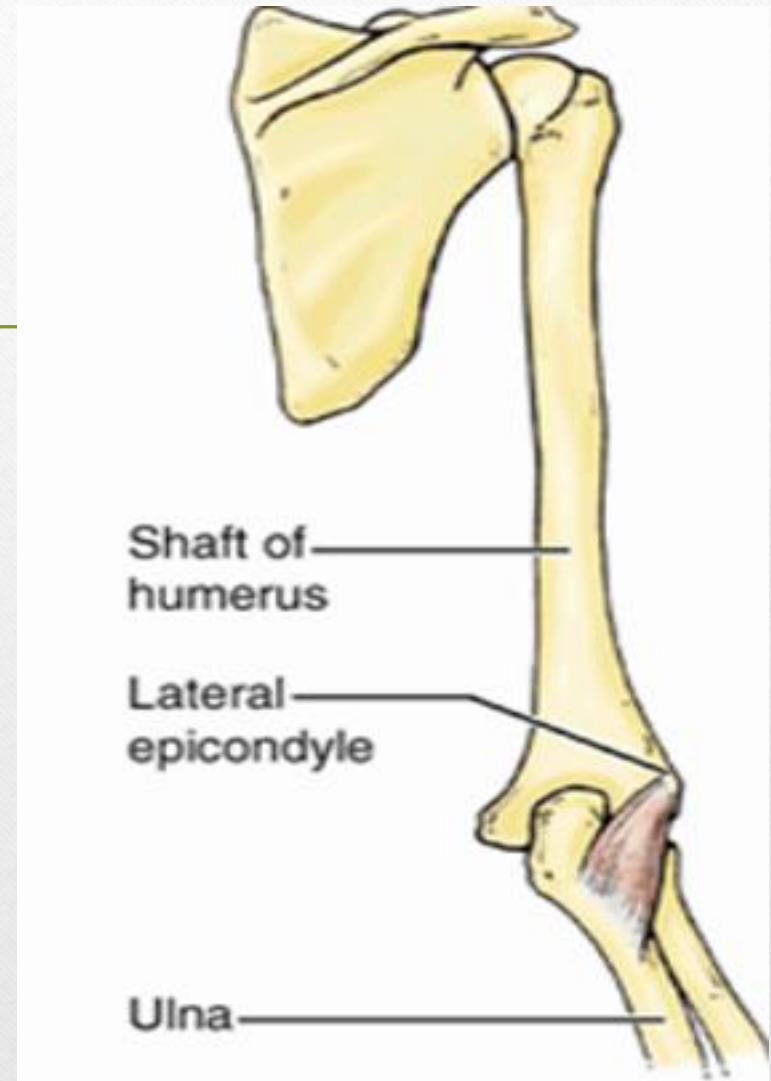
Triceps brachii

- **Origin:**
 - **Long head:** infra-glenoid tubercle of scapula
 - **Lateral head:** posterior surface of humerus, superior to radial groove
 - **Medial head:** posterior surface of humerus, inferior to radial groove
- **Insertion:** Proximal end of olecranon of ulna and fascia of forearm
- **Innervation:** Radial nerve
- **Action:** Chief extensor of forearm; long head resists dislocation of humerus



Anconeus

- **Origin:** Lateral epicondyle of humerus
- **Insertion:** Lateral surface of olecranon and superior part of posterior surface of ulna
- **Innervation:** Radial nerve
- **Action:** Assists triceps in extending forearm; stabilizes elbow joint; may abduct ulna during pronation

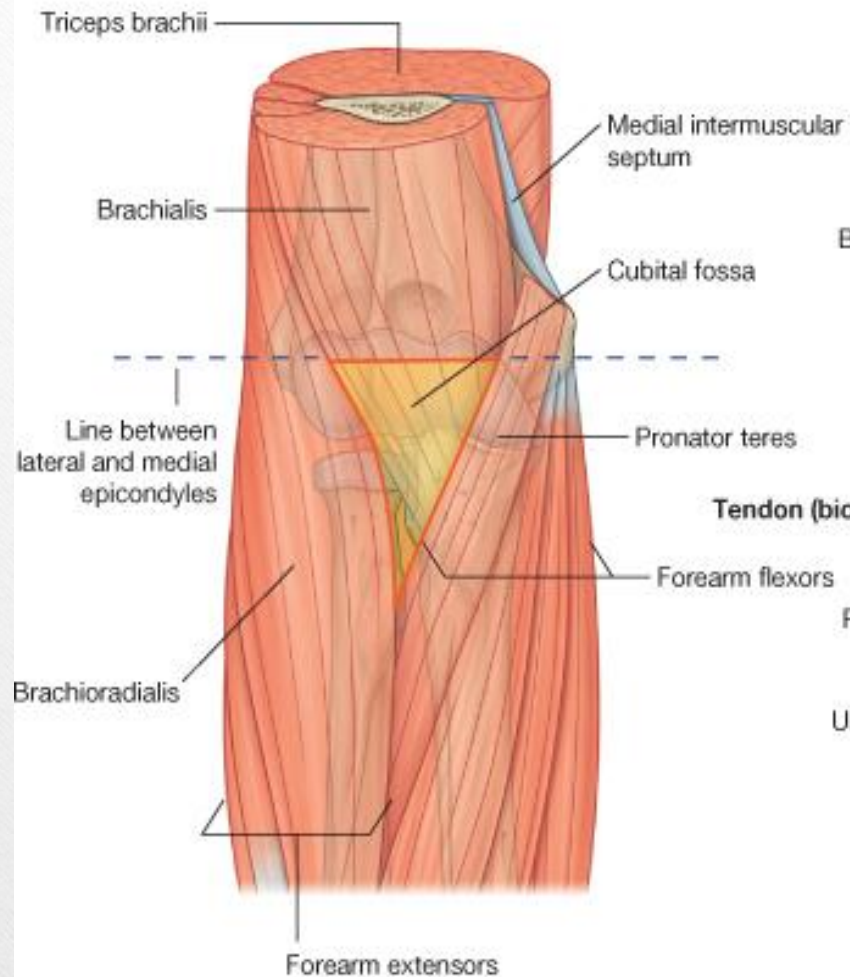
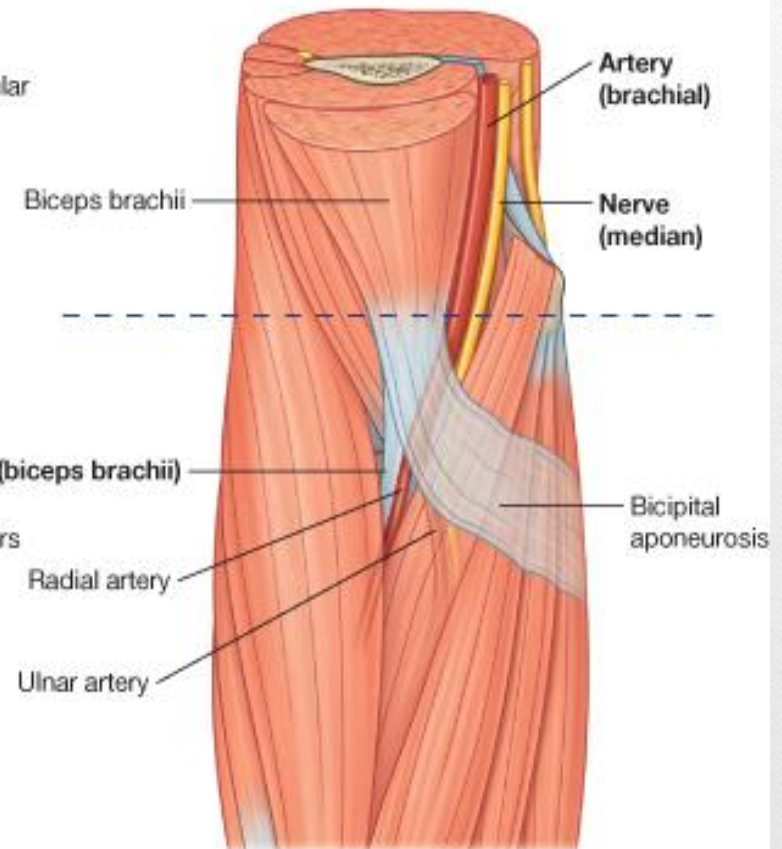


Cubital fossa

A triangular depression on anterior surface of the elbow

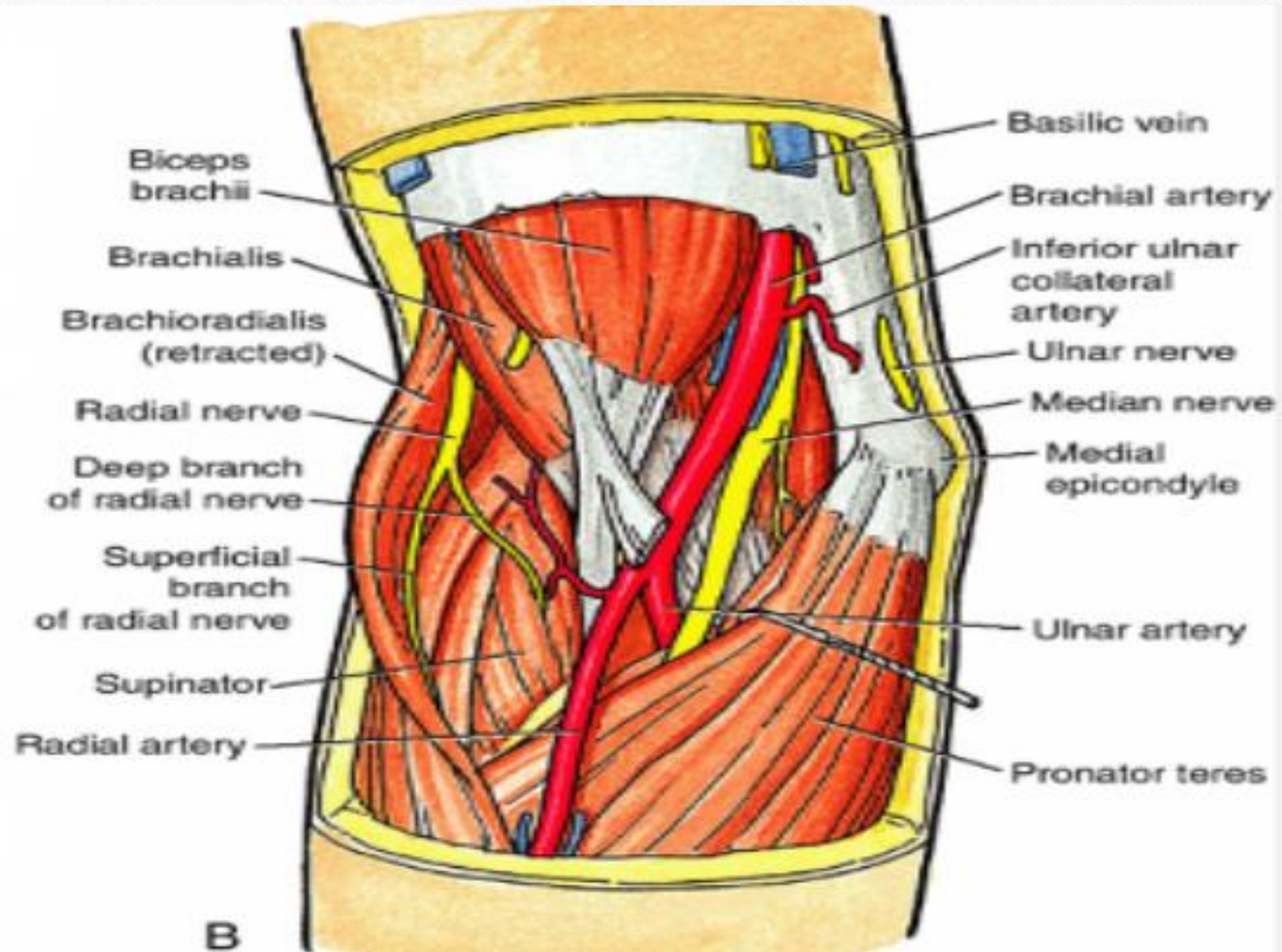
Boundaries

- **Superiorly**;- an imaginary line connecting the medial and lateral epicondyles
- **Medially**:- pronator teres
- **Laterally**:- brachioradialis
- **Floor**:- brachialis and supinator muscles
- **Roof** :-the continuity of brachial and antebrachial (deep) fascia, reinforced by the bicipital aponeurosis, subcutaneous tissue and skin

A**B**

Contents

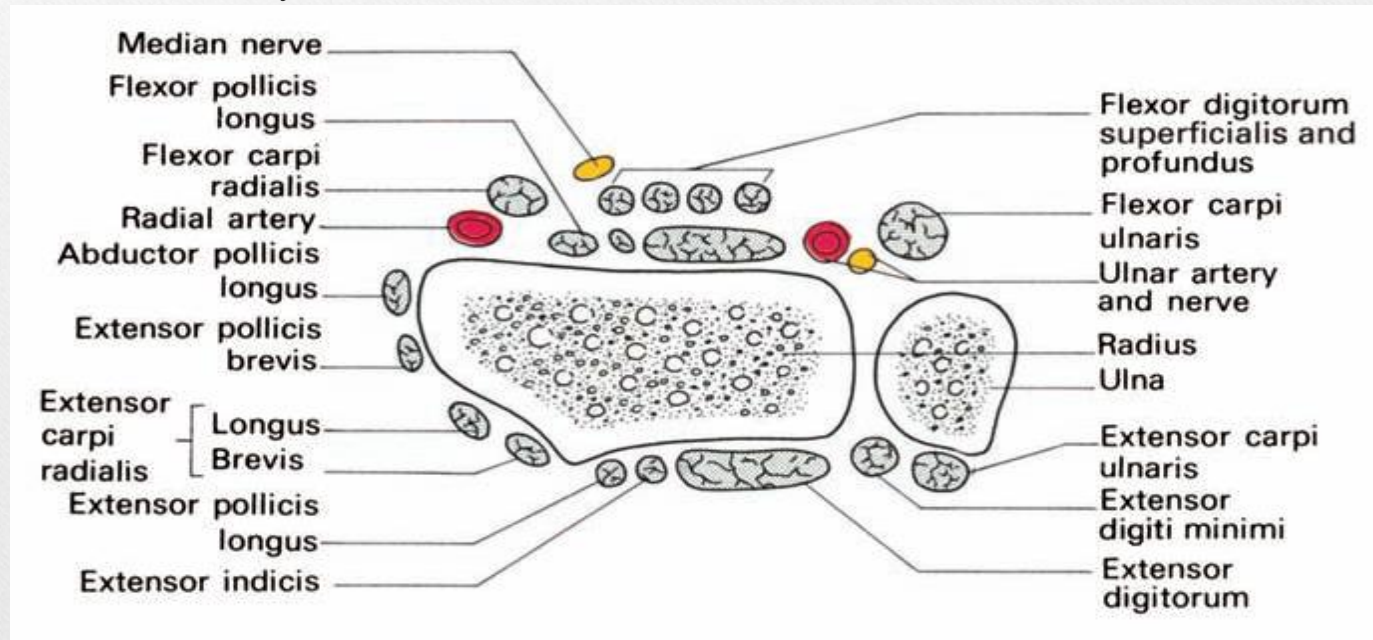
- Brachial artery and branches, radial and ulnar arteries
 - Accompanying veins of the arteries
-
- Biceps brachii tendon
 - Median nerve
 - Radial nerve, dividing into its superficial and deep branches
 - Median cubital vein, medial and lateral cutaneous nerves of the forearm- on the subcutaneous tissue overlying the cubital fossa



Forearm

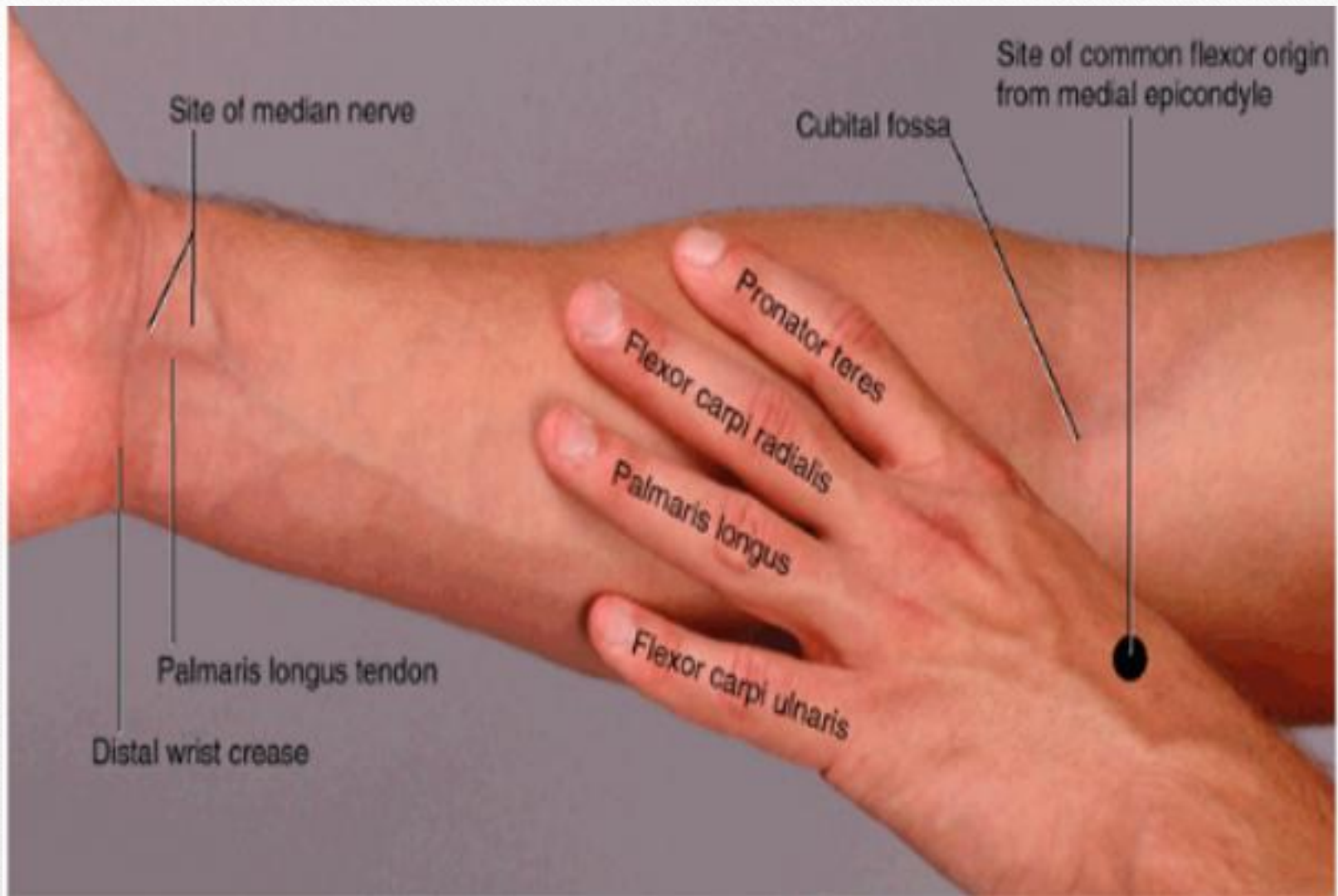
Muscles of Forearm

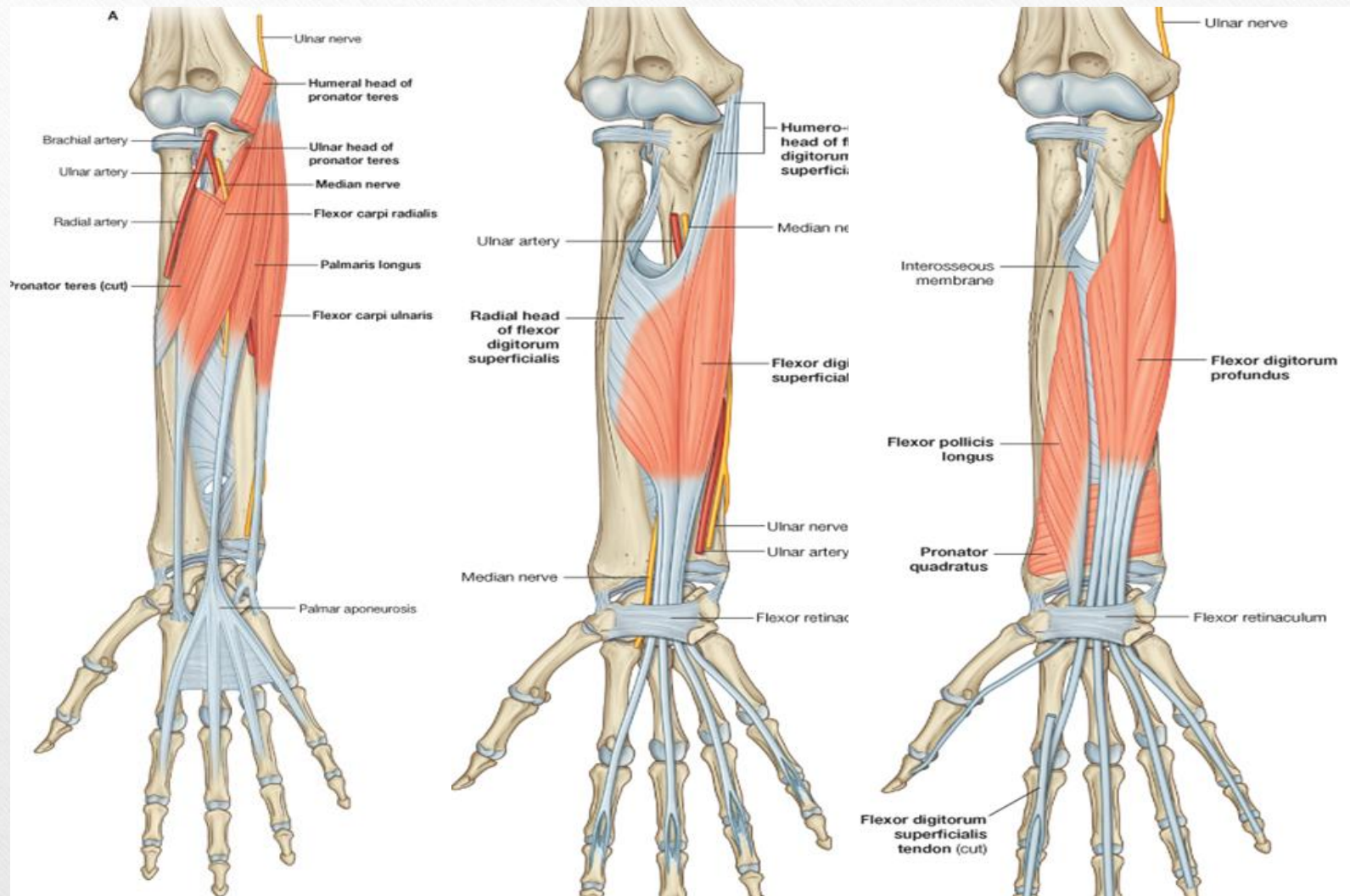
- Anterior compartment:- **flexors - pronators** mm, and are served mainly by the **median nerve**
- Posterior compartment:- **extensor - supinators** mm and innervated by the **radial nerve**



Anterior compartment muscles

- Are arranged in three layers
- **Superficial layers:-**
 - Have common origin- a flexor tendon on medial epicondyle of humerus
 - Innervated by median nerve except flexor carpi ulnaris
 - **Pronator teres:-** Inserted on middle of lateral surface of radius & is pronator and flexor of the forearm
 - **Flexor carpi radialis:-** Inserted on bases of 2nd metacarpal, it flexes and abduct hand
 - **Palmaris longus:-** insertion is on distal half of flexor retinaculum ; it flex hands
 - **Flexor carpi ulnaris:-** inserted in to pisiform, hook of hamate, 5th metacarpal bones ; it flexes and abduct hand and is innervated by ulnar nerve

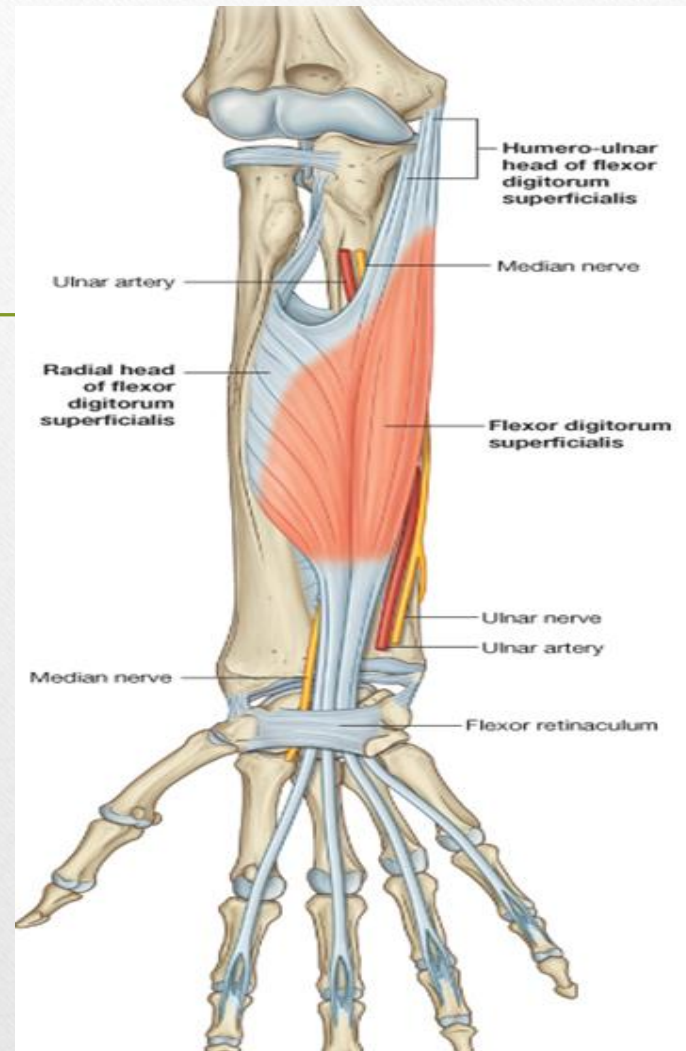




➤ Intermediate layer

■ **Flexor digitorum**

superficialis :- arise from the common origin and inserted in to middle phalanges of medial four fingers; served by median nerve and is flexor of middle phalanges



➤ Deep layer

Flexor digitorum profundus:-

- Arise from proximal $\frac{3}{4}$ of ulna and interosseous membrane and inserted to bases of distal phalanges
- Served by ulnar and median nerves; is flexor of distal phalanges

Flexor pollicis longus

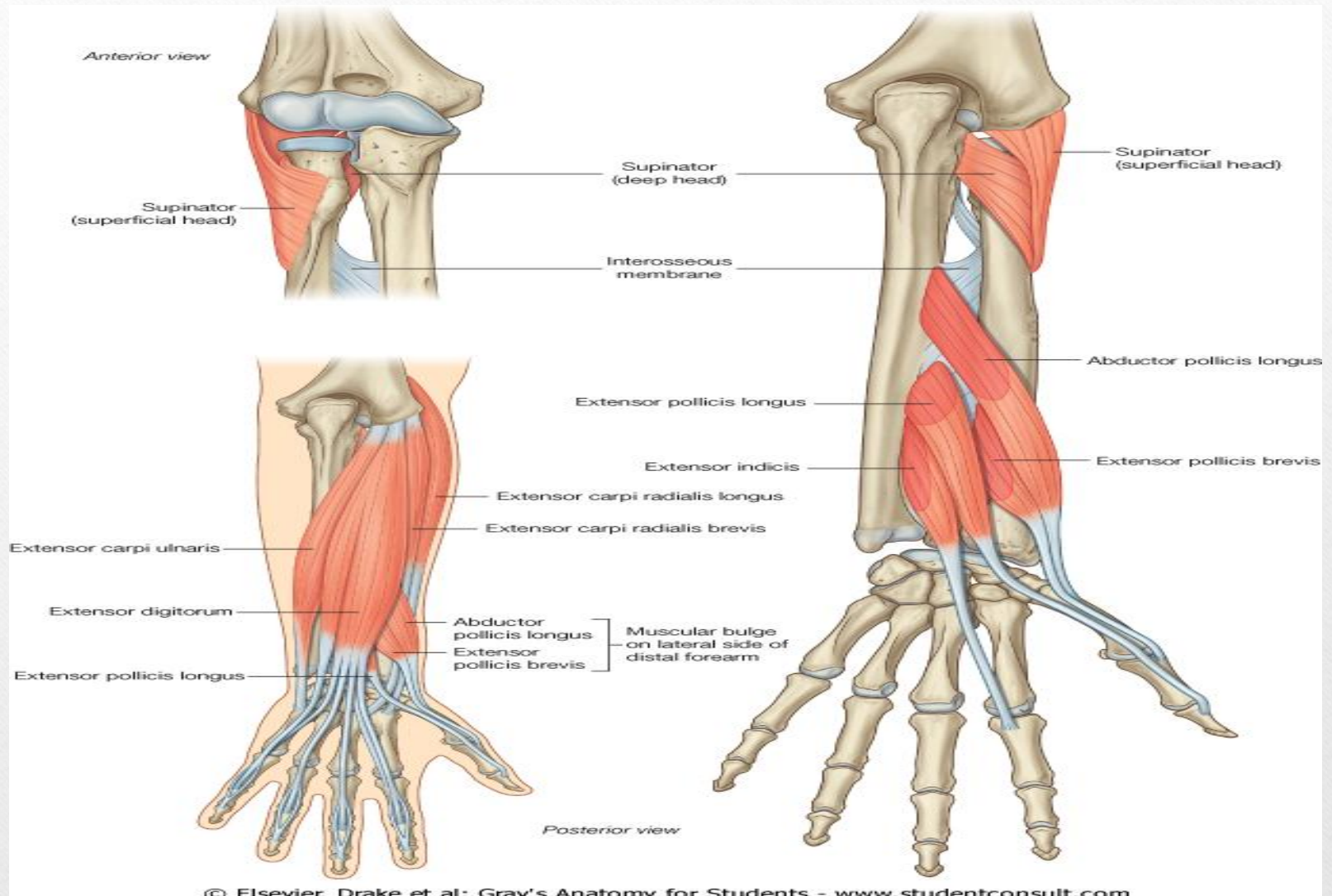
- Arise from anterior surface of radius and interosseous membrane and inserted in to base of distal phalanx of thumb
- It flex the thumb and innervated by **median nerve**

Pronator quadratus

- Originated from distal quarter of anterior surface of ulna and end on distal quarter of anterior surface of radius/Median nerve

Muscles in the posterior compartment of the forearm

- All are innervated by radial nerve
- **Superficial extensors**
 - Attached proximally by a common extensor tendon to lateral epicondyle
 - Extensor carpi radialis brevis
 - Extensor digitorum, Extensor digiti minimi
 - Extensor carpi ulnaris, Brachioradialis
 - Extensor carpi radialis longus
- **Deep extensors**
 - Abductor pollicis longus, Extensor pollicis brevis, Extensor pollicis longus, Extensor indicis



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Superficial layer

- **Brachioradialis**

- Arise from supraepicondylar ridge of humerus and Inserted in to Lateral surface of distal end of radius proximal to styloid process
-

- Functionally is flexor of the forearm

- **Extensor carpi radialis longus and brevis**

- Originated from lateral supraepicondylar ridge and epicondyle of humerus and inserted in to base of 2nd and 3rd metacarpal bones
- They extend and abduct hand at wrist joint

- **Extensor (digitorum & digiti minimi) muscle**

- Extend from common extensor origin to extensor expansions of fingers
- They extends fingers at metacarpophalangeal and interphalangeal joints

- **Extensor carpi ulnaris muscle**

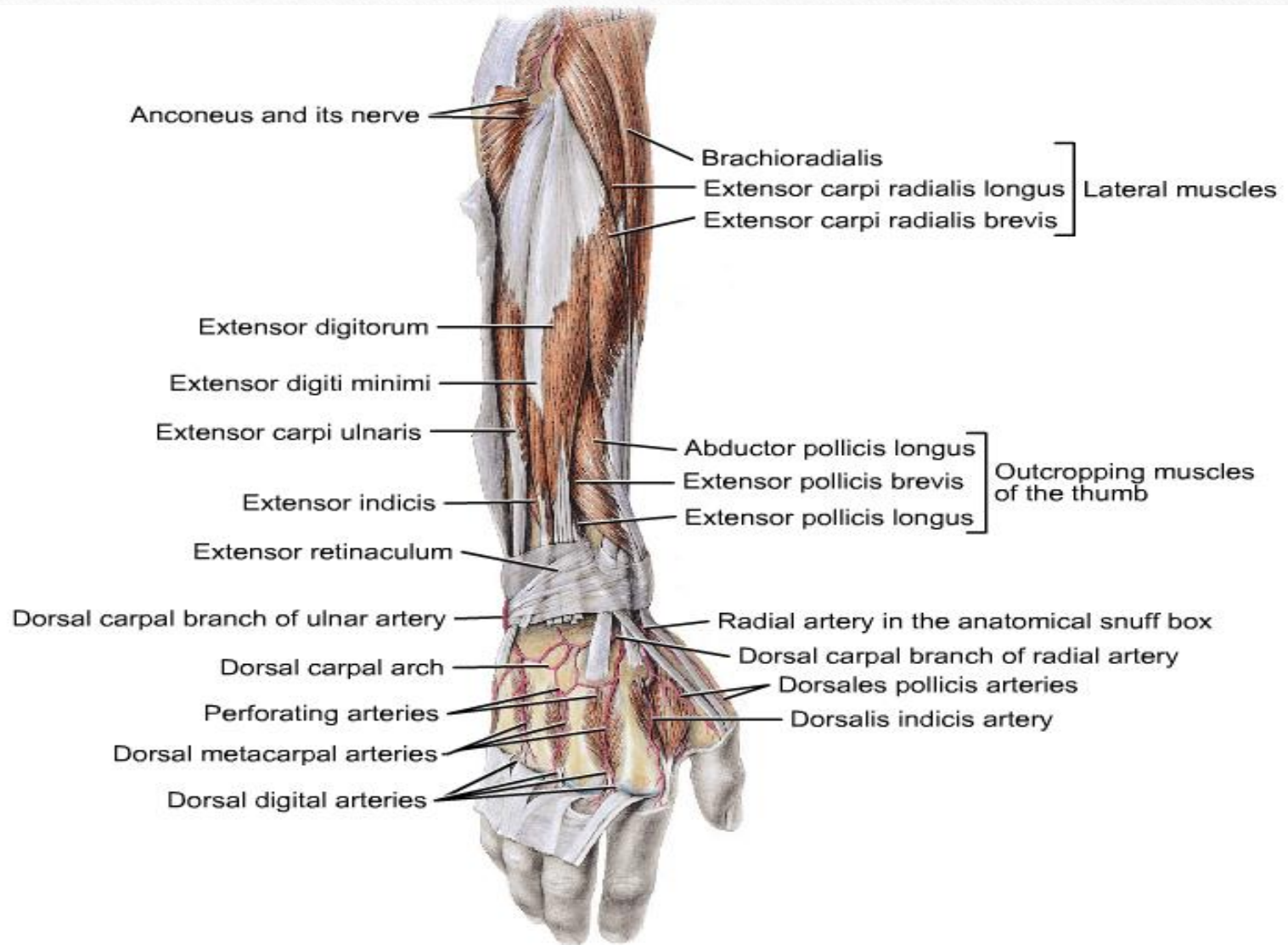
Deep layer

➤ Supinator

- Extend from common origin to proximal 3rd of radius
- Supinator of the forearm

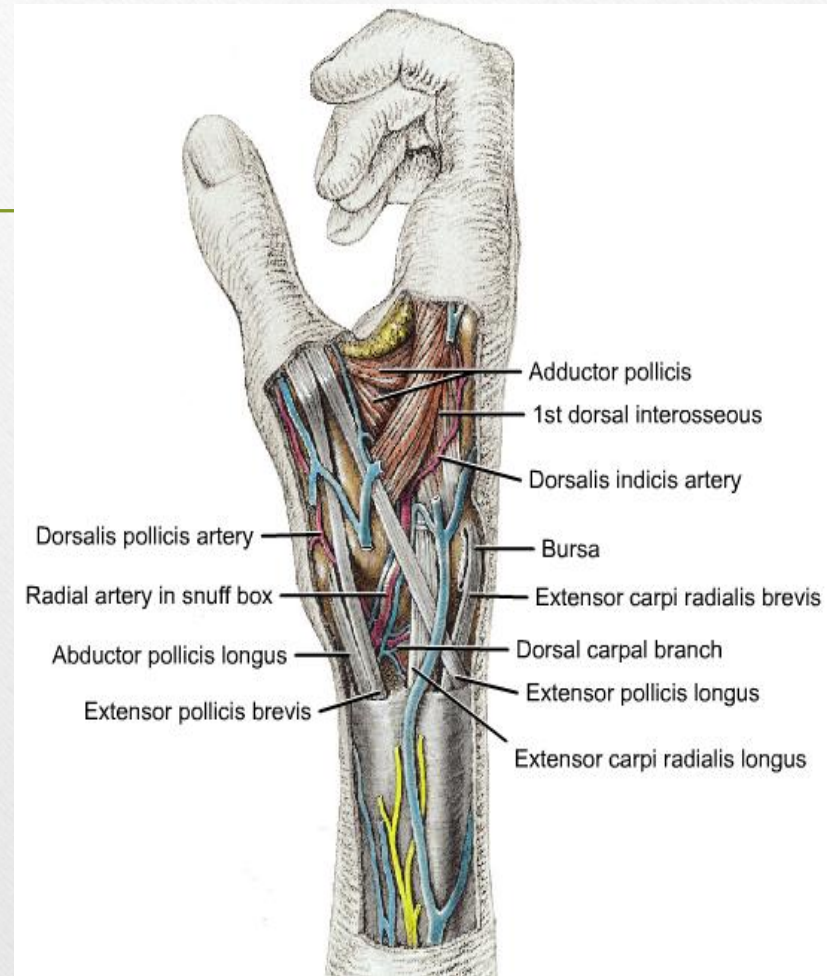
➤ Outcropping muscles

- **Abductor pollicis longus**
- **Extensor pollicis longus**
- **Extensor pollicis brevis**
 - They extend from the posterior surface of ulna and radius up to phalanx of the thumb
 - Are extensor and abductor of the phalanx of the thumb
- **Extensor indicis**
 - It covers the area b/n distal third of ulna to extensor expansion of 2nd finger ; it extends 2nd finger = “**pointing position**”



Anatomical snuff box

- A triangular shallow depression on the lateral aspect of the wrist when the thumb is extended fully
- The tendons of the **abductor pollicis longus** and **extensor pollicis brevis** bound anatomical snuff box laterally, and tendon of **extensor pollicis longus** bounds it medially
- **Radial artery** lies on the floor of the snuffbox



The hand

➤ Fascia of palm:-

- **Palmar aponeurosis**
- Strong condensation of parts of the deep fascia of the palm
- Covers the soft tissues and overlies the flexor tendons
- Apex is continuous with the flexor retinaculum and palmaris longus tendon
- Four longitudinal digital bands radiate from the apex and attach distally to the bases of the proximal phalanges



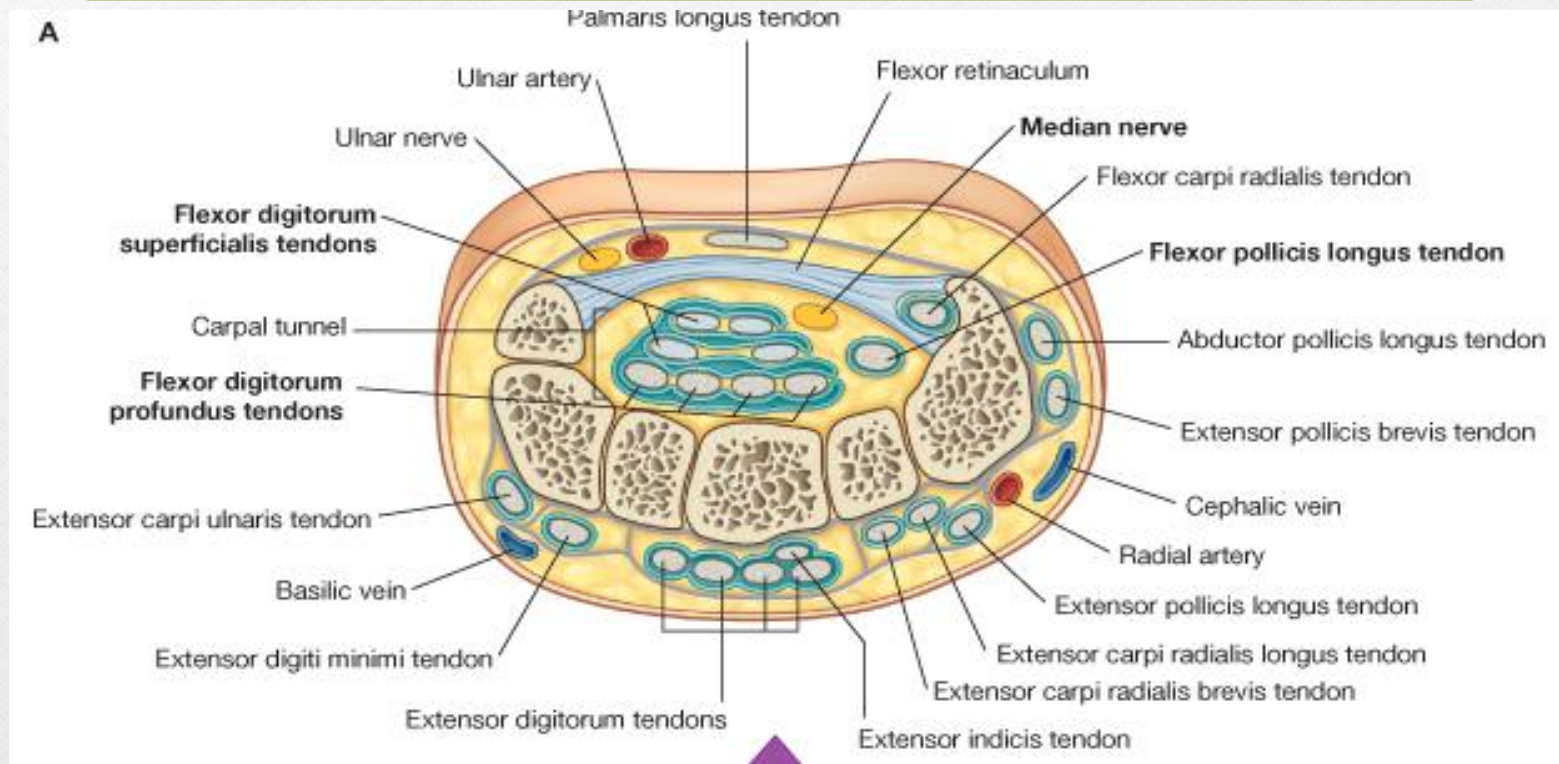
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Flexor retinaculum

- Fibrous band covering anterior concavity of carpus and converts it into a **carpal tunnel**
- Attached in to medially to **pisiform and hook of hamate** and laterally to **scaphoid** and **trapezium**
- Relations
 - Structures passing **superficial** to the retinaculum: **tendon of palmaris longus, palmar cutaneous of medial and ulnar nerves, ulnar vessels and ulnar nerve**
 - Structures passing **deep** to the retinaculum: **medial nerve, tendons of flexor digitorum superficialis, profundus, flexor pollicis longus**

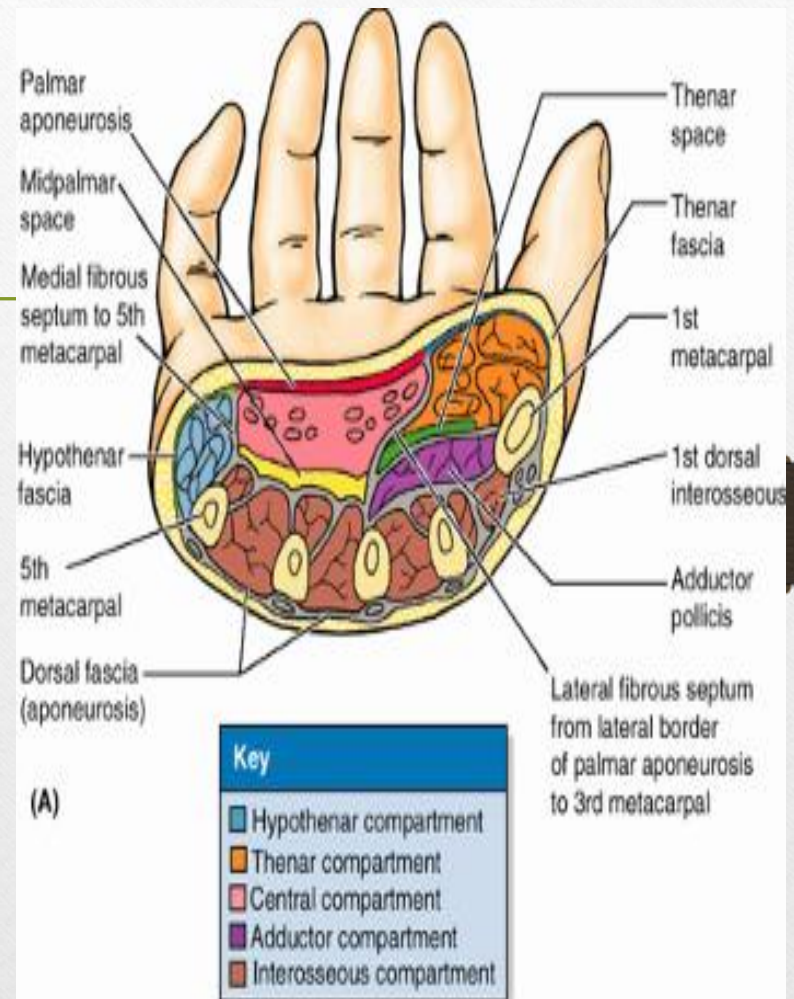
Carpal Tunnel Syndrome:-

- Reduces the size of the carpal tunnel
- **Median nerve** is the mostly affected
- Progressive loss of coordination and strength in the thumb



Muscles of the hand

- The intrinsic muscles of the hand are located in five compartments
 - **Thenar muscles** in thenar compartment: ~~abductor pollicis brevis~~, flexor pollicis brevis, and opponens pollicis
 - **Hypothenar muscles** in hypothenar compartment: abductor digiti minimi, flexor digiti minimi brevis and opponens digiti minimi
 - **Adductor pollicis** in adductor compartment
 - **Lumbricals** in central compartment
 - **Interossei** in interosseous compartment

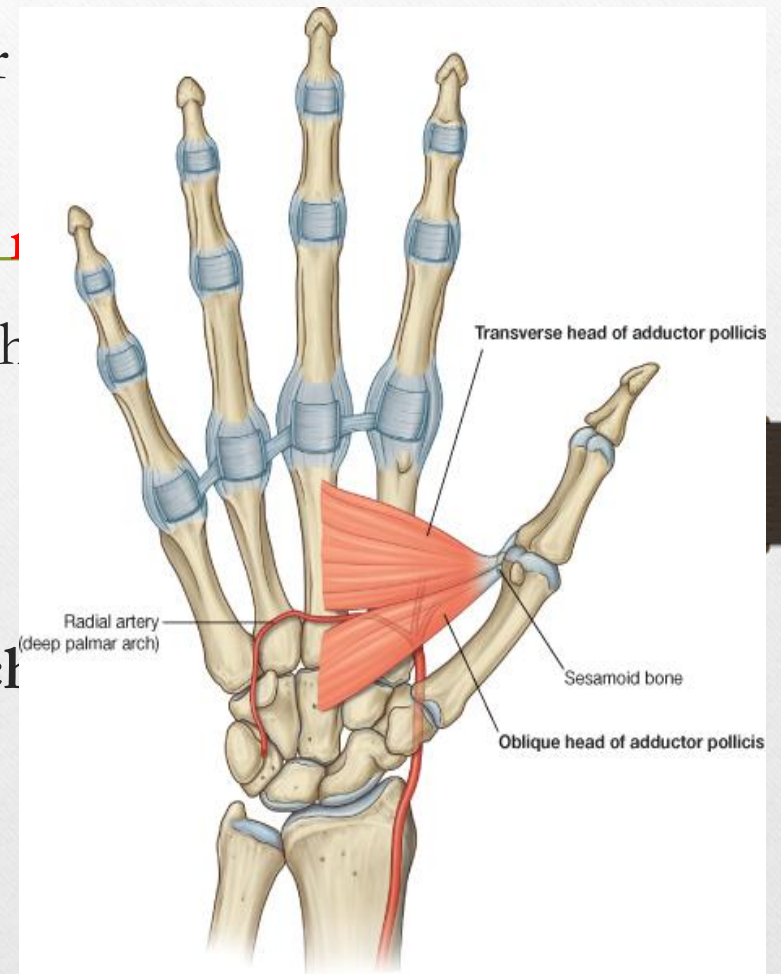


- **Thenar muscles**

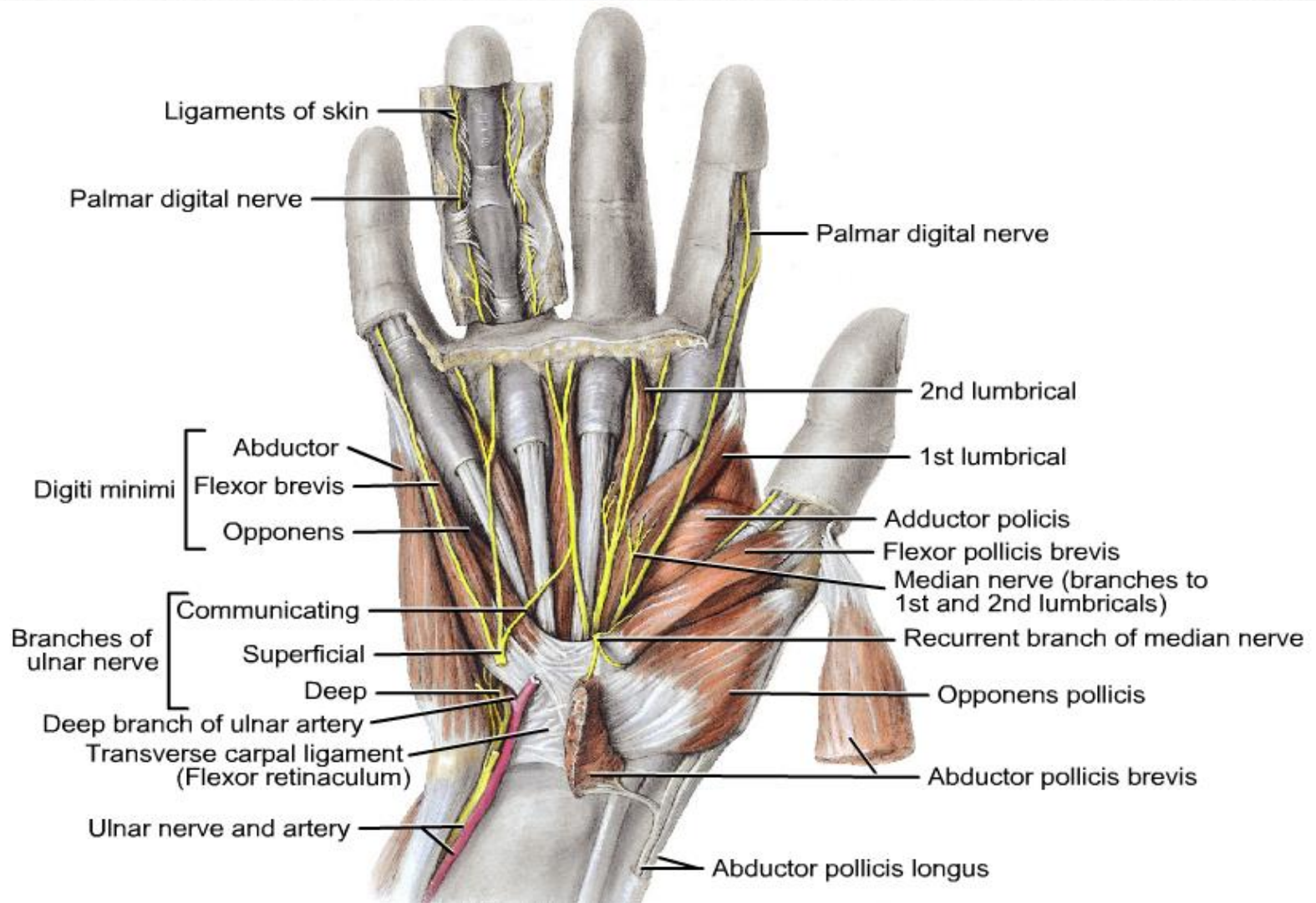
- Abductor pollicis brevis, flexor opponens pollicis
- Served by recurrent branch of 1
- They oppose abduct and flex th

- **Adductor pollicis**

- It adduct the thumb
- Innervated by deep branch

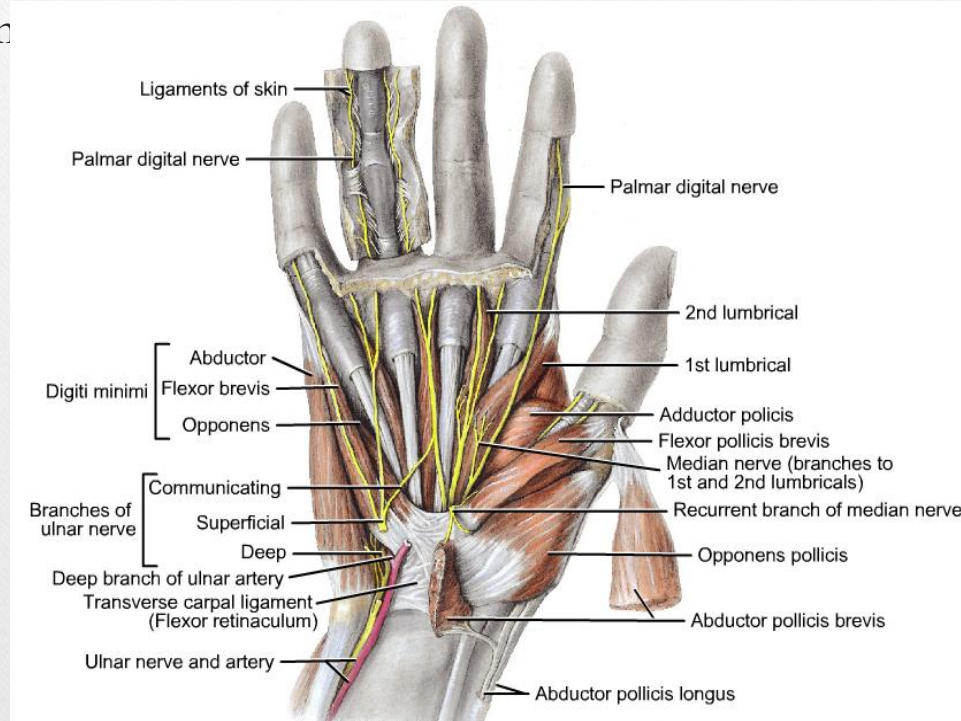


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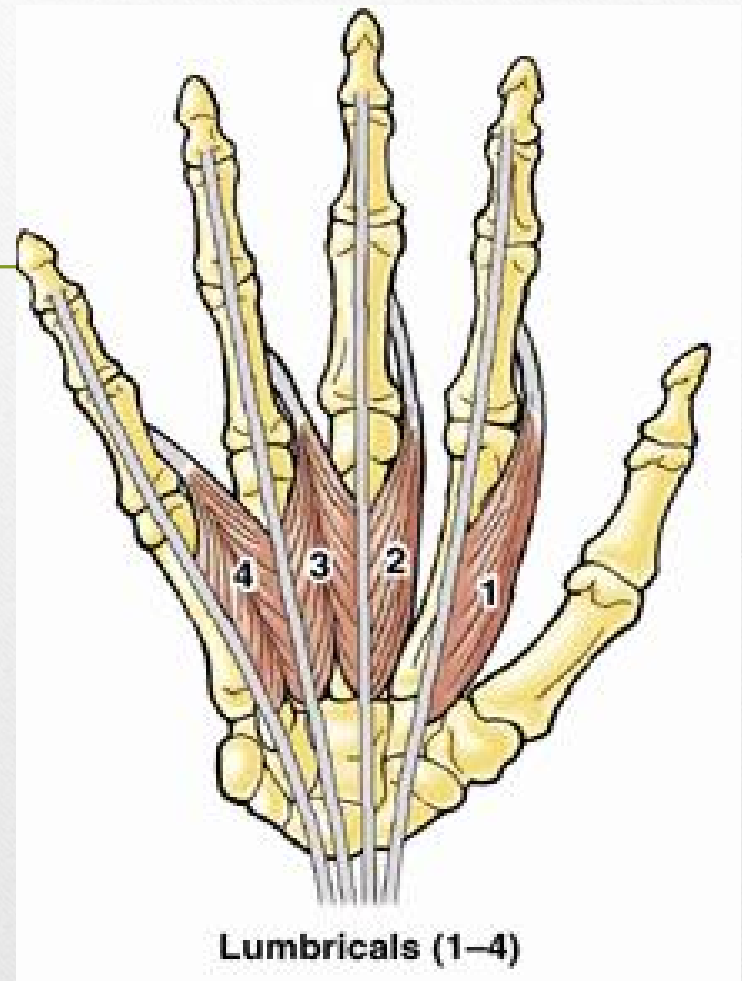
Hypothenar muscles

- Abductor digiti minimi, flexor digiti minimi brevis and opponens digiti minimi
- Supplied by deep branch of the **ulnar nerve**
- Act on the sn



Lumbricals- 1st to 4th

- Attached in to 2nd to 5th fingers and metacarpal bones
- **Innervation:**
 - 1st and 2nd: **Median nerve**
 - 3rd and 4th: Deep branch of **ulnar nerve**
- **Action: Flex** metacarpophalangeal joints; **extend** interphalangeal joints of 2nd - 5th fingers



Interossei

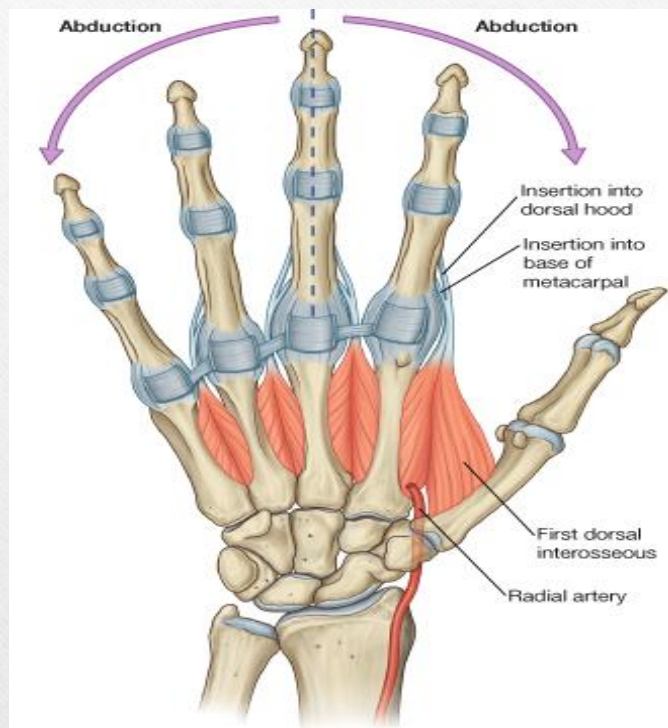
- **Dorsal**

-
- **Attached** to metacarpals and 2nd to 4th fingers
 - **Supplied** by deep branch of **ulnar nerve**
 - **Abduct** 2nd to 4th fingers

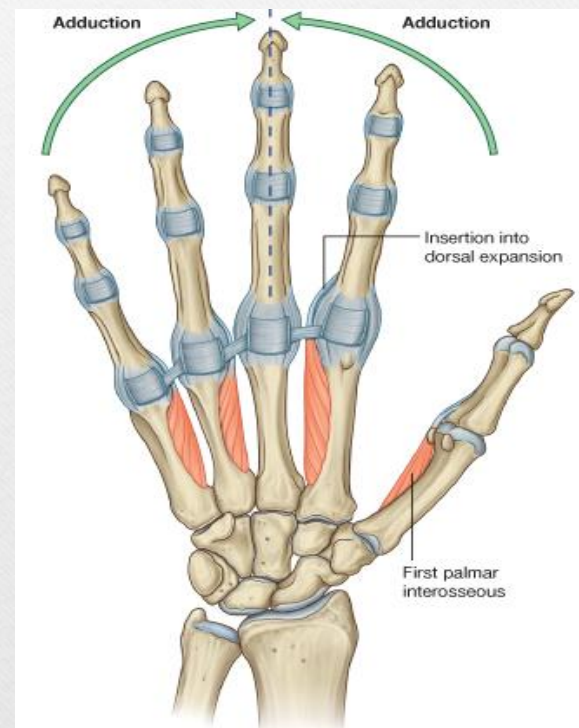
- **Palmar**

- **Attached on to** 2nd, 4th, and 5th metacarpals and fingers
- **Supplied** by deep branch of **ulnar nerve**
- **Adduct** 2nd, 4th, and 5th fingers

Dorsal interossei



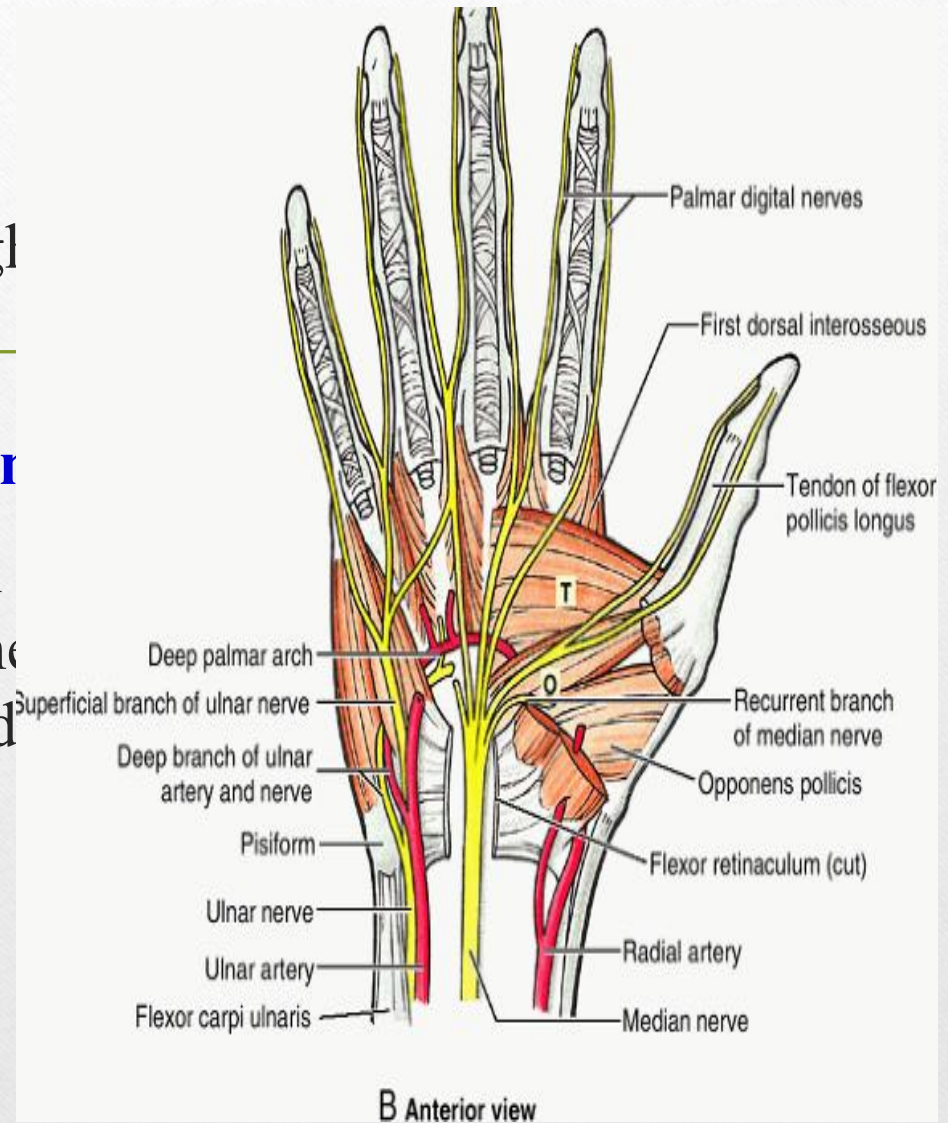
Palmar interossei

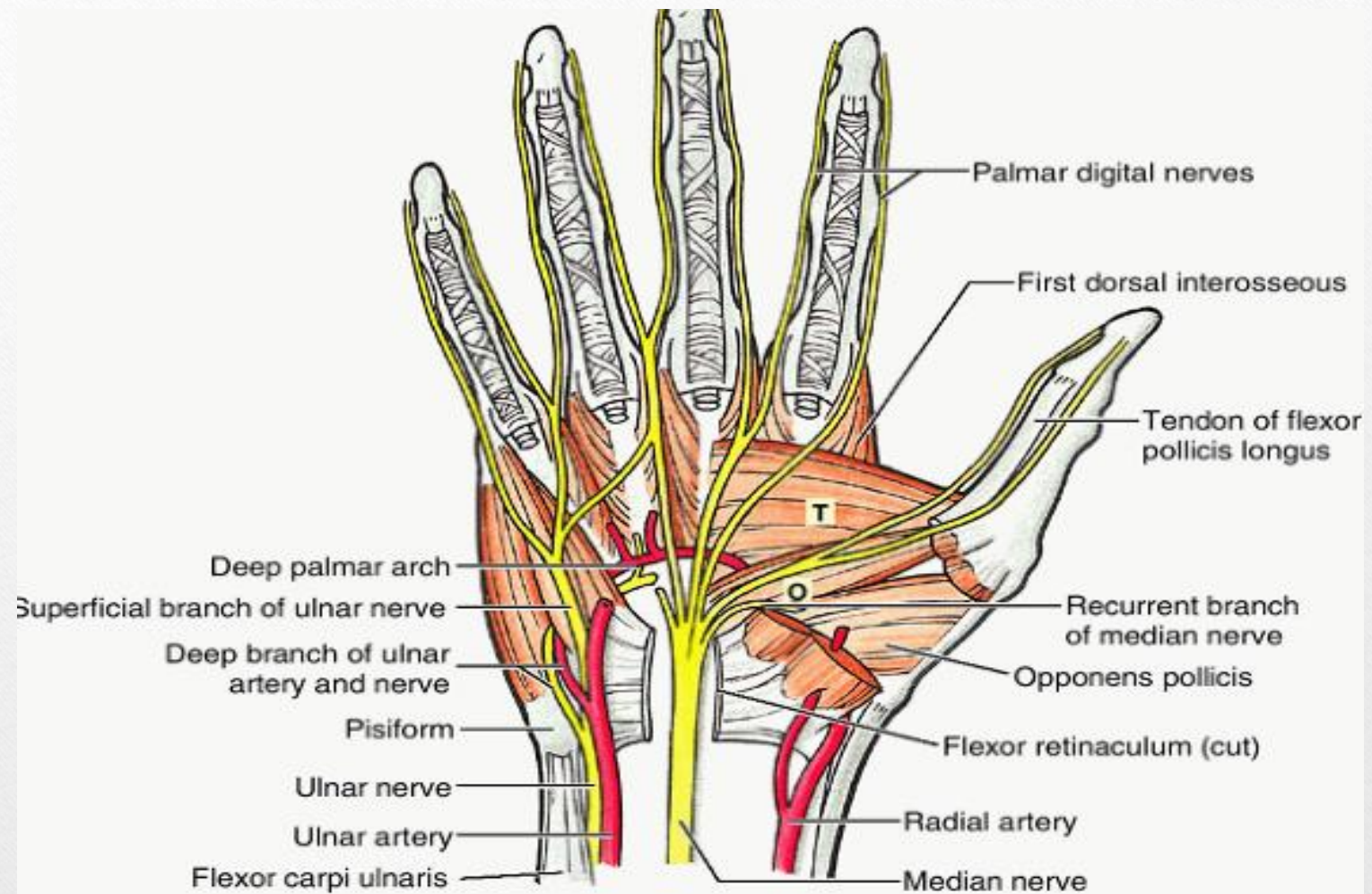


Nerves of hand

Median nerve

- Enters the hand through
- Supplies
 - **Thenar muscles** and
 - Skin on the lateral first three digits, the the dorsum of the d





B Anterior view

Ulnar nerve

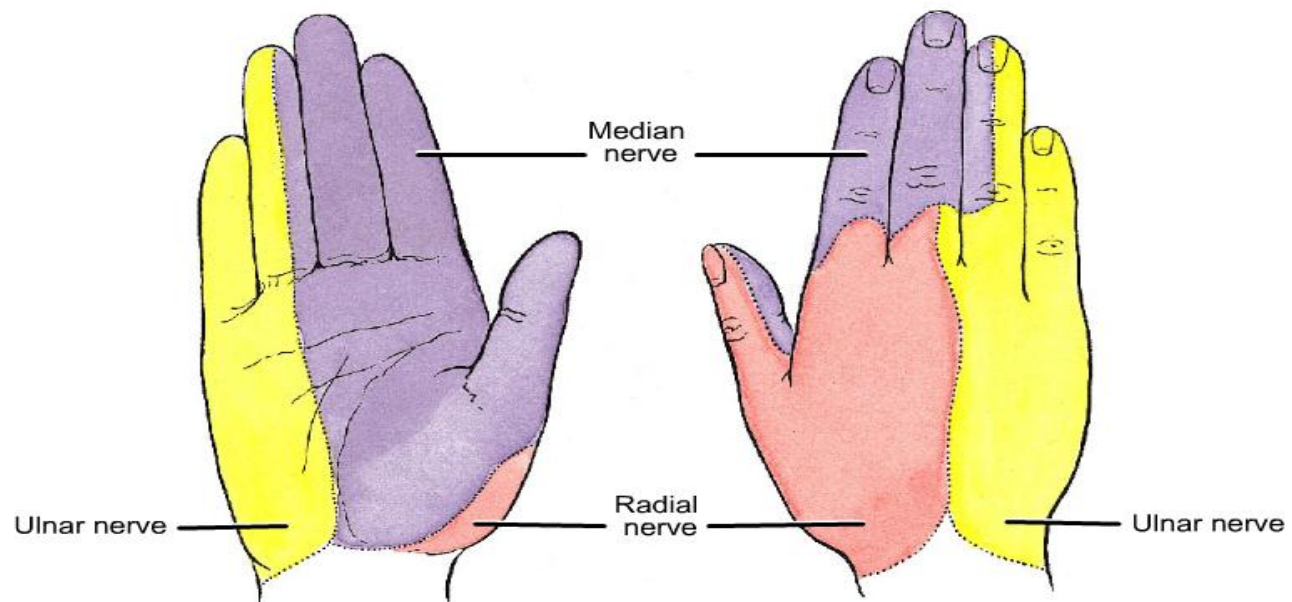
- Leaves the forearm by emerging from **deep to the tendon** of the **flexor carpi ulnaris**
- Just proximal to the wrist, the ulnar nerve gives off
 - **palmar cutaneous branch**

 - supplies skin on the medial side of the palm
 - **dorsal cutaneous branch**
 - supplies the medial half of dorsum of the hand, the 5th finger, and the medial half of the 4th finger
- Ends at the flexor retinaculum by dividing into 2 branches
 - **superficial branch**: supplies anterior surface of medial one and a half fingers
 - **deep branch**: supplies **hypothenar** muscles, **medial two lumbricals**, **adductor pollicis**, and all the **interossei**

Radial nerve

- Supplies no hand muscles
- Supply the skin and fascia over the lateral two thirds

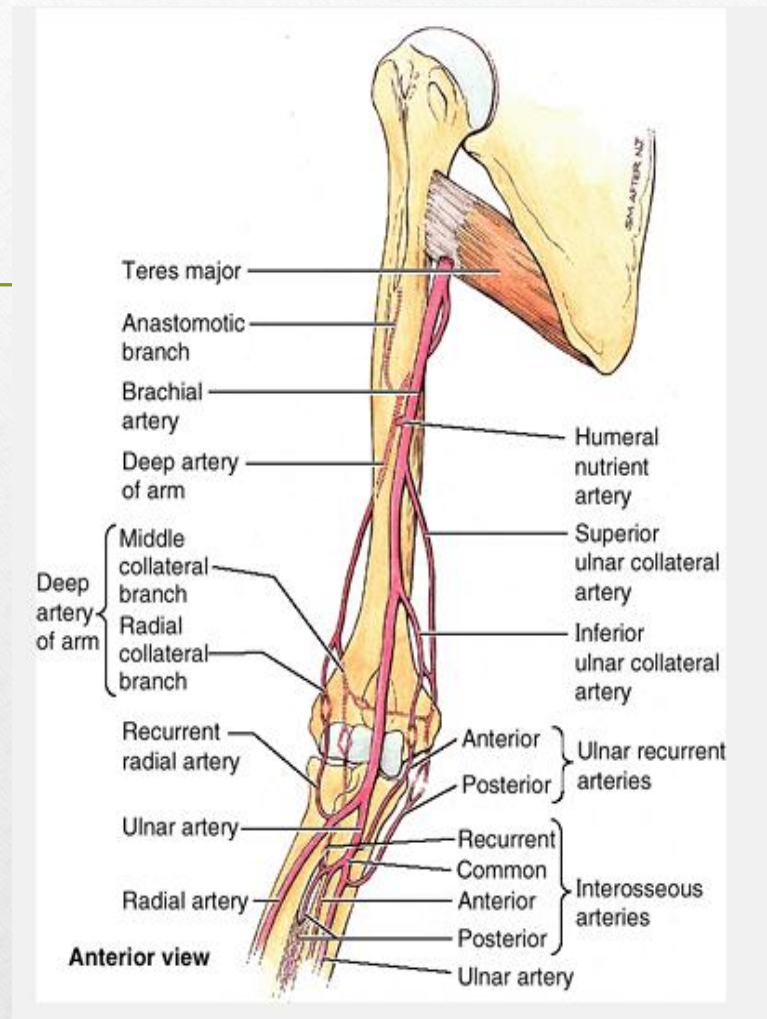
of the
proxim



Arteries of the upper limb

Brachial artery

- Provides the main arterial supply to the arm
- Is the continuation of the axillary artery
- Begins at the inferior border of teres major
- Ends in the cubital fossa opposite the neck of the radius by dividing into **radial** and **ulnar** arteries

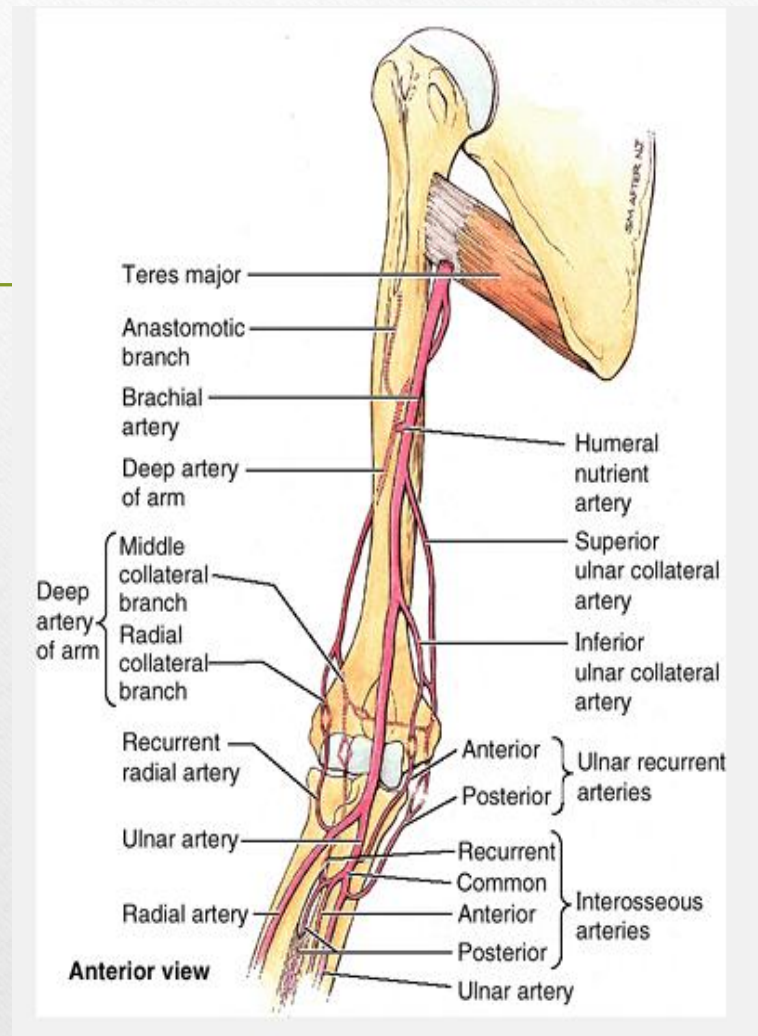


Arteries of the upper limb

Brachial artery

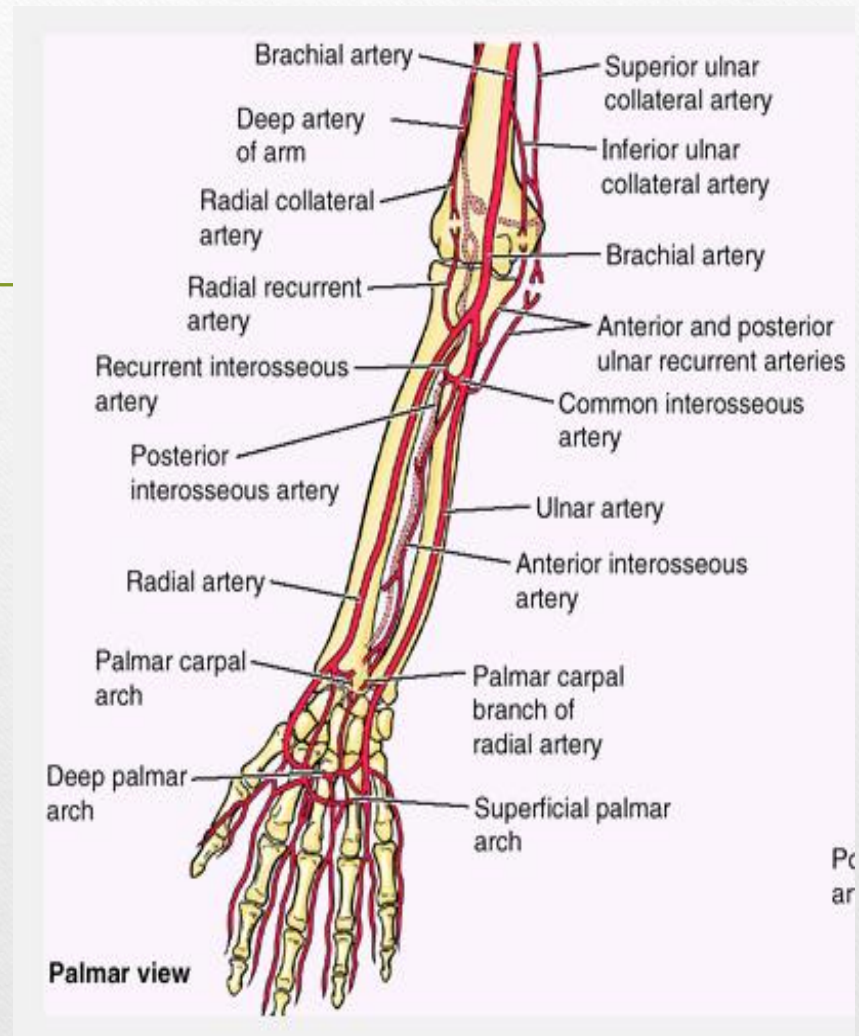
• Branches

- unnamed muscular branches
- humeral nutrient artery
- deep artery of arm / profunda brachii
- superior and inferior ulnar collateral arteries
- **Radial** and **ulnar** arteries



Artery of the forearm

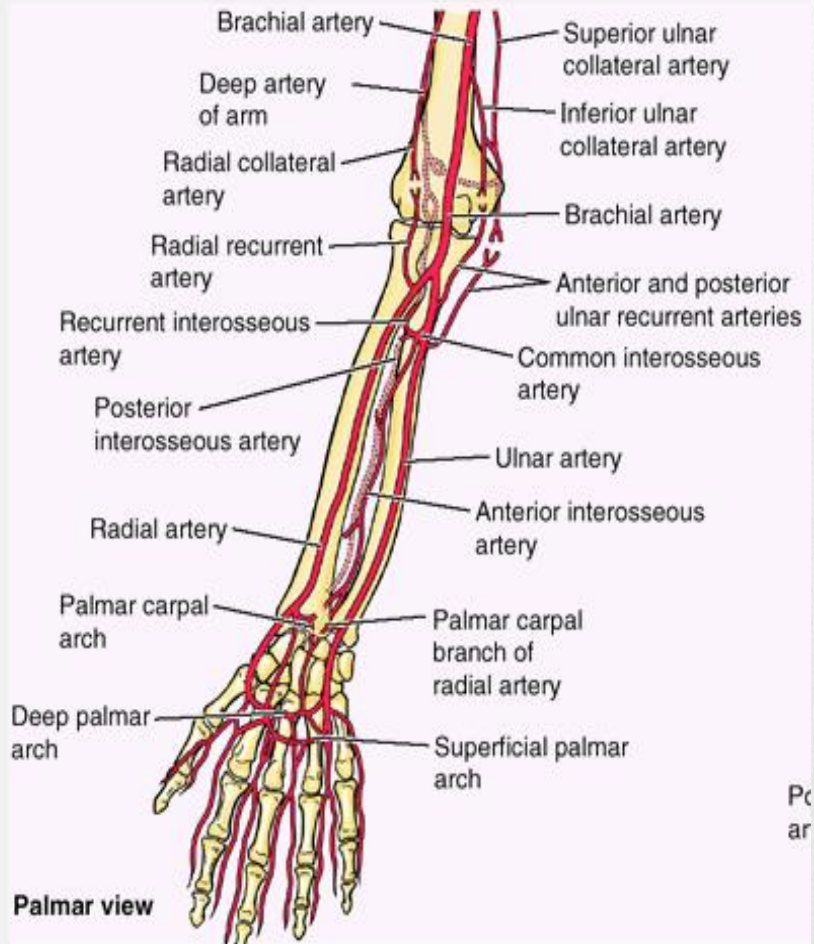
- Ulnar aa- from brachial aa
- Branches
 - Anterior and Posterior ulnar **recurrent** artery
 - Common interosseous artery
 - Anterior and Posterior interosseous artery
 - Recurrent interosseous artery
 - Palmar carpal branch
 - Dorsal carpal branch



Artery of the forearm con't

Radial aa:- branch of brachial artery

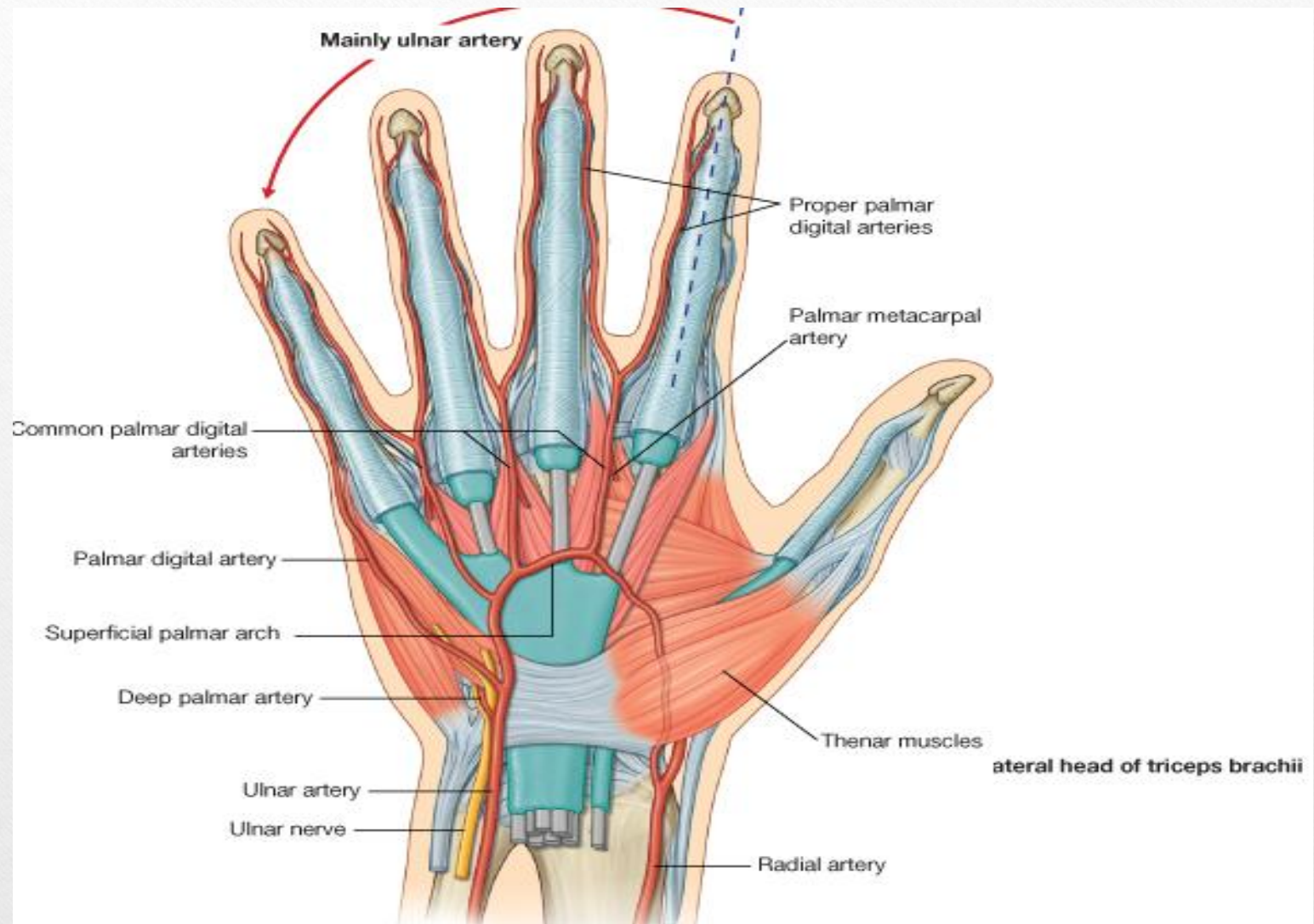
- **Branches**
 - Radial recurrent artery
 - Palmar carpal branch
 - Dorsal carpal branch



Arteries of Hand

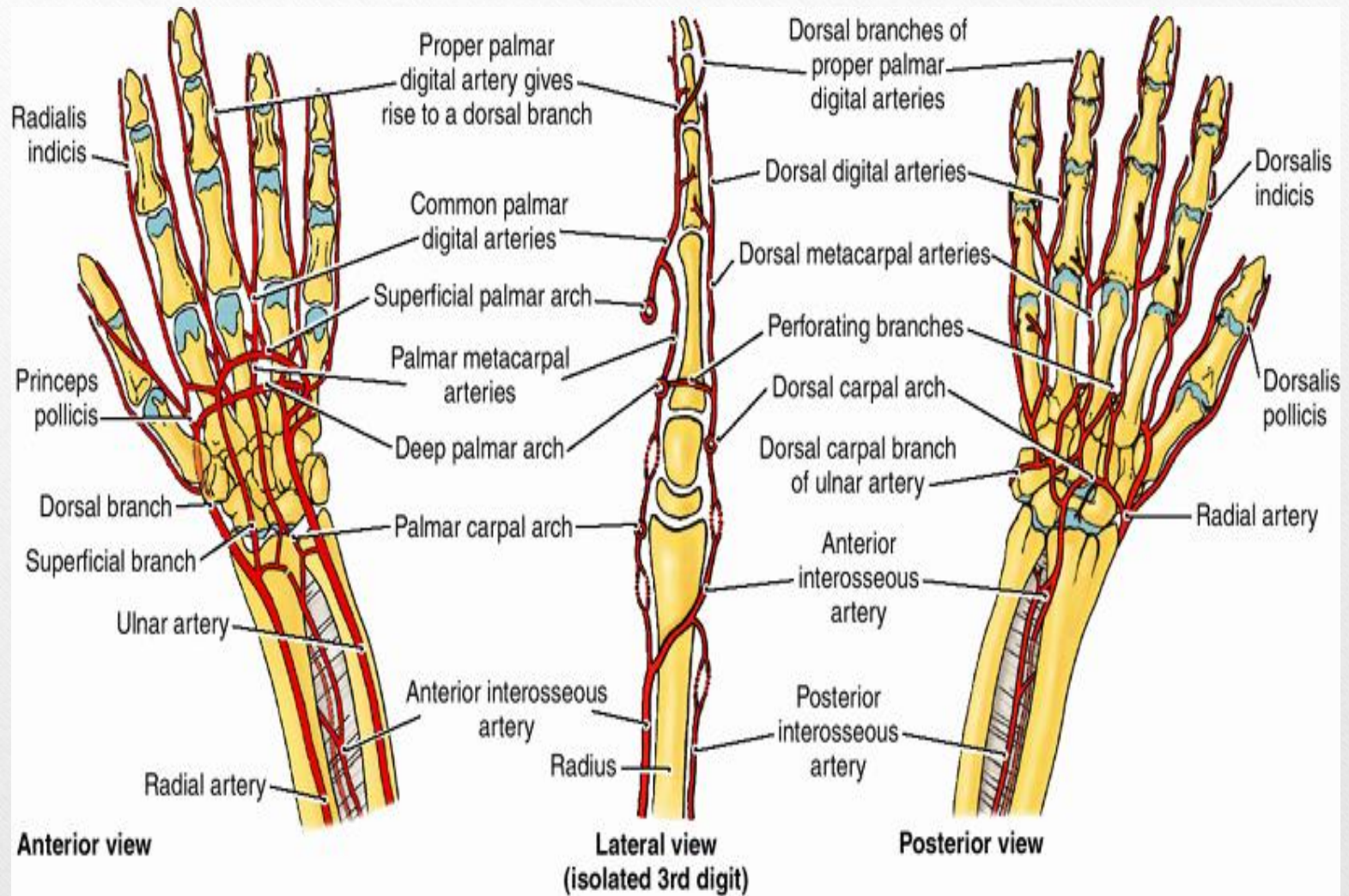
Ulnar artery

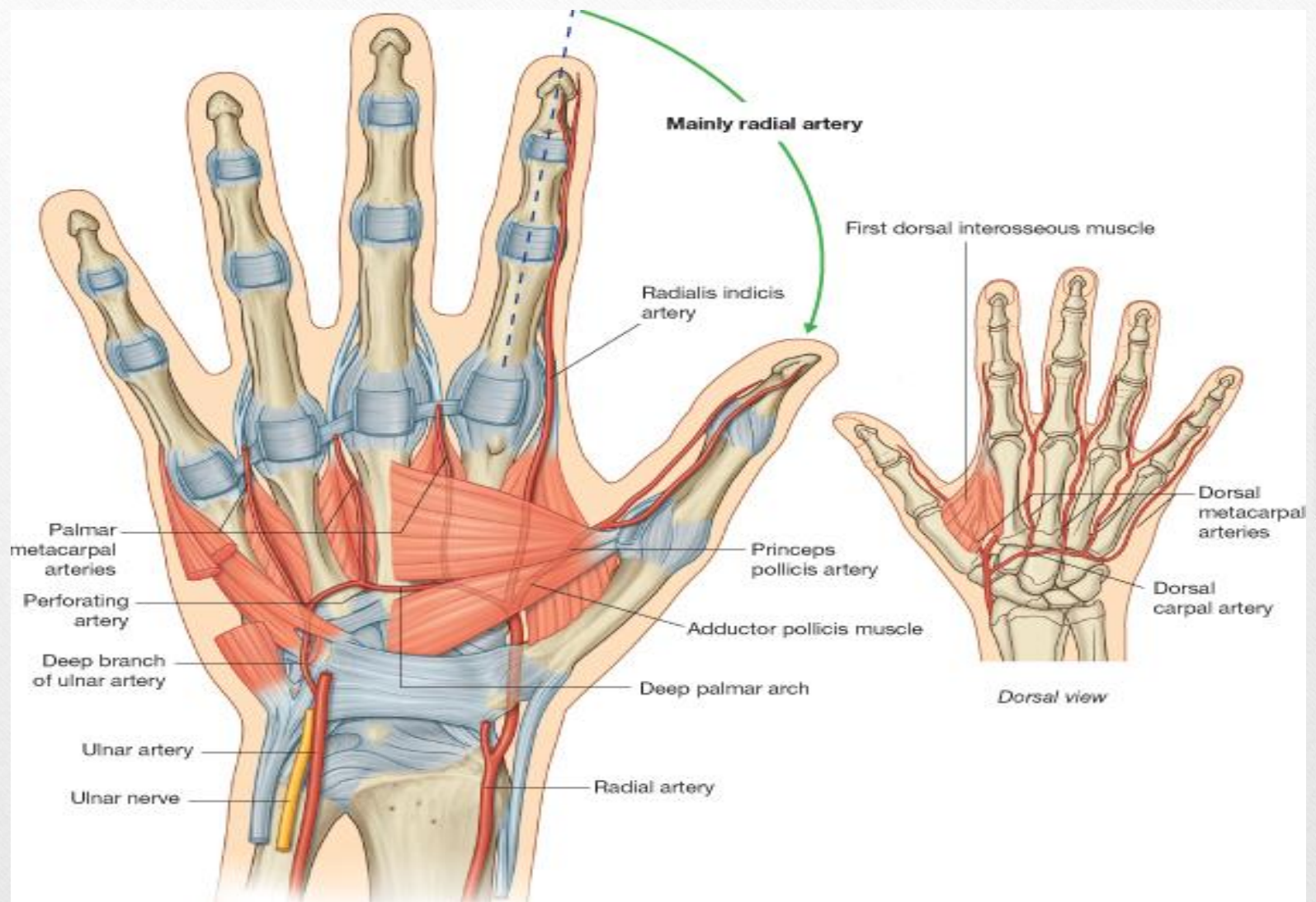
- Enters the hand anterior to the flexor retinaculum between the pisiform and the hook of hamate via the ulnar canal (Guyon canal)
- Lies **lateral to the ulnar nerve**
- Divides into two terminal branches
 - **Superficial palmar arch**
 - **Deep palmar branch**
- Superficial palmar arch
 - main termination of the ulnar artery
 - gives rise to three **common palmar digital** arteries that anastomose with palmar metacarpal arteries from the deep palmar arch
 - Each common palmar digital artery divides into a pair of **proper palmar** digital arteries that run along the adjacent side of the 2nd-4th fingers



Radial artery

- Curves dorsally around the scaphoid and trapezium in the floor of anatomical snuff box
- Enters the palm by passing between the heads of 1st dorsal interosseous muscle
- Turns medially and passes between the heads of the adductor pollicis
- Ends by anastomosing with the deep branch of the ulnar artery to form the deep palmar arch
- **Deep palmar arch**
 - formed mainly by **radial artery**
 - lies across the metacarpals just distal to their bases
 - gives rise to three **palmar metacarpal arteries**, **princeps pollicis** artery and **radialis indicis artery**

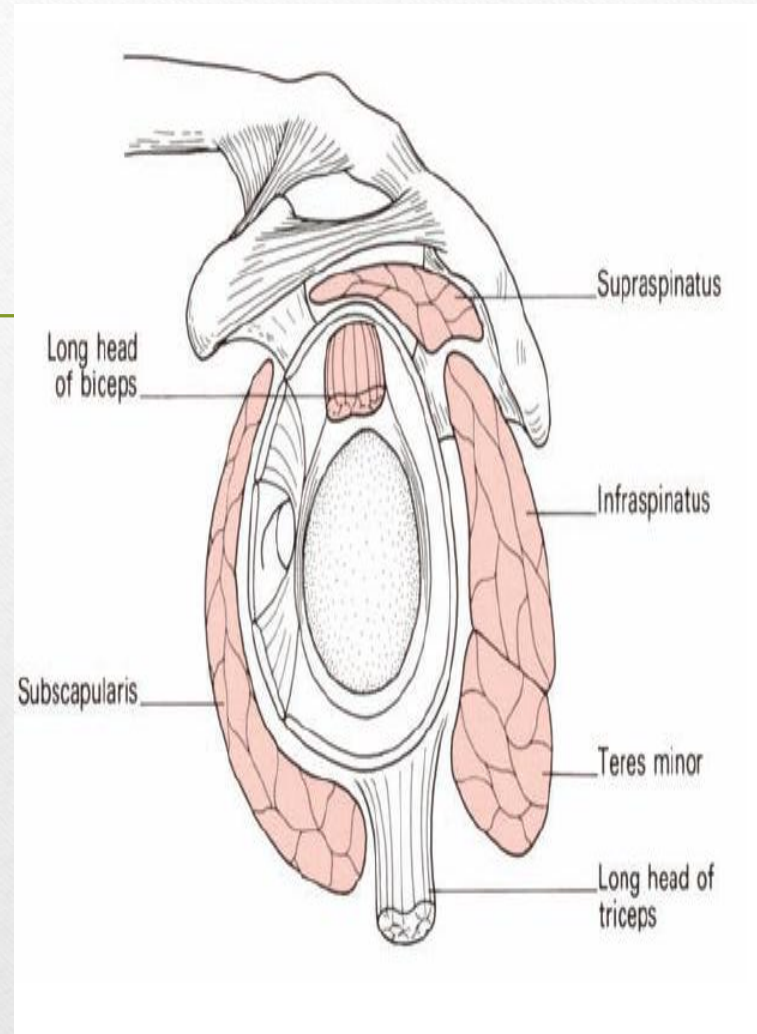




Joints of Upper Limb

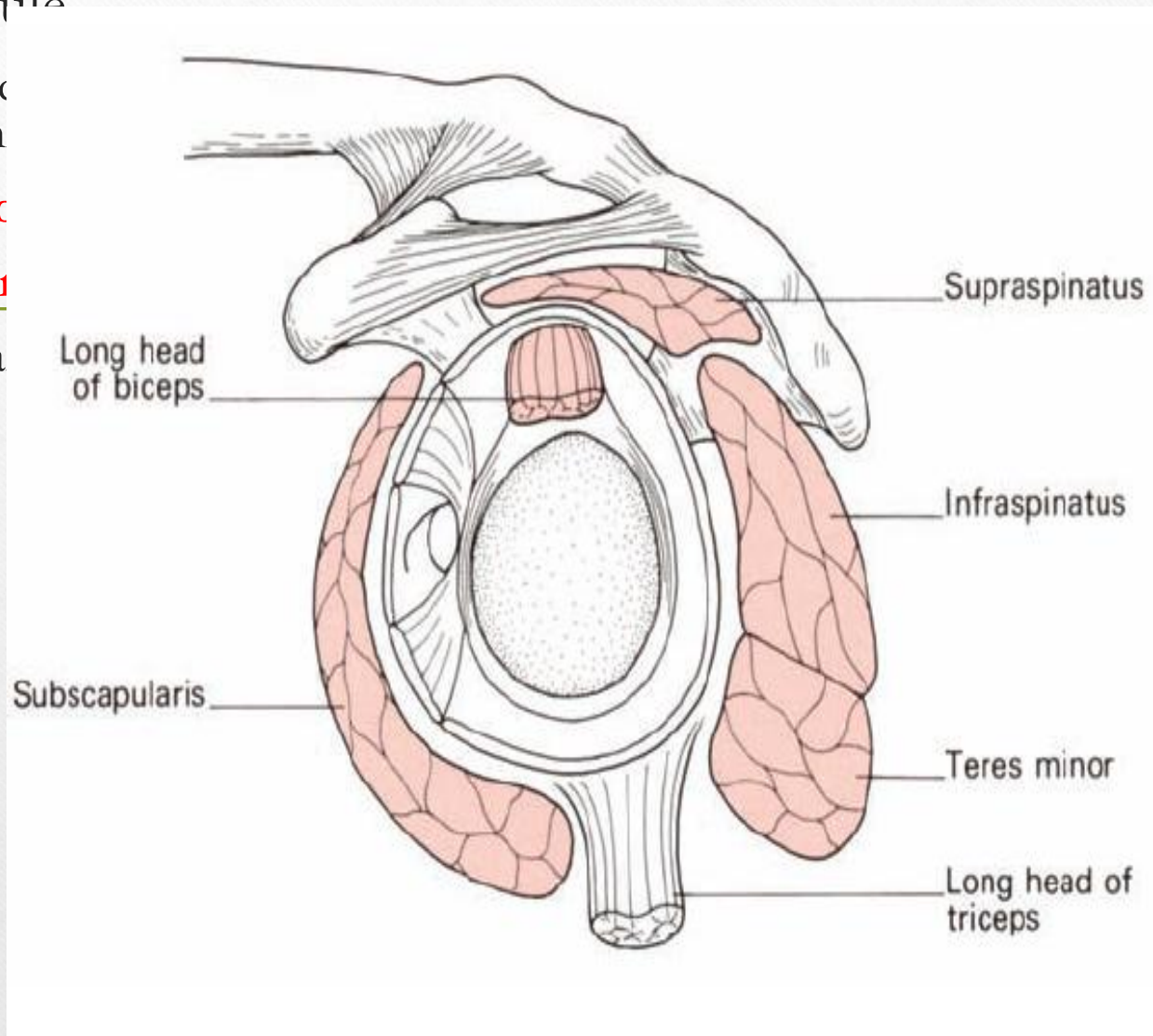
Glenohumeral Joint

- Type: **ball and socket synovial joint**
 - permits a wide range of movement
- Articulation
 - The humeral head articulates with shallow glenoid cavity of the scapula held in the cavity by musculotendinous rotator cuff

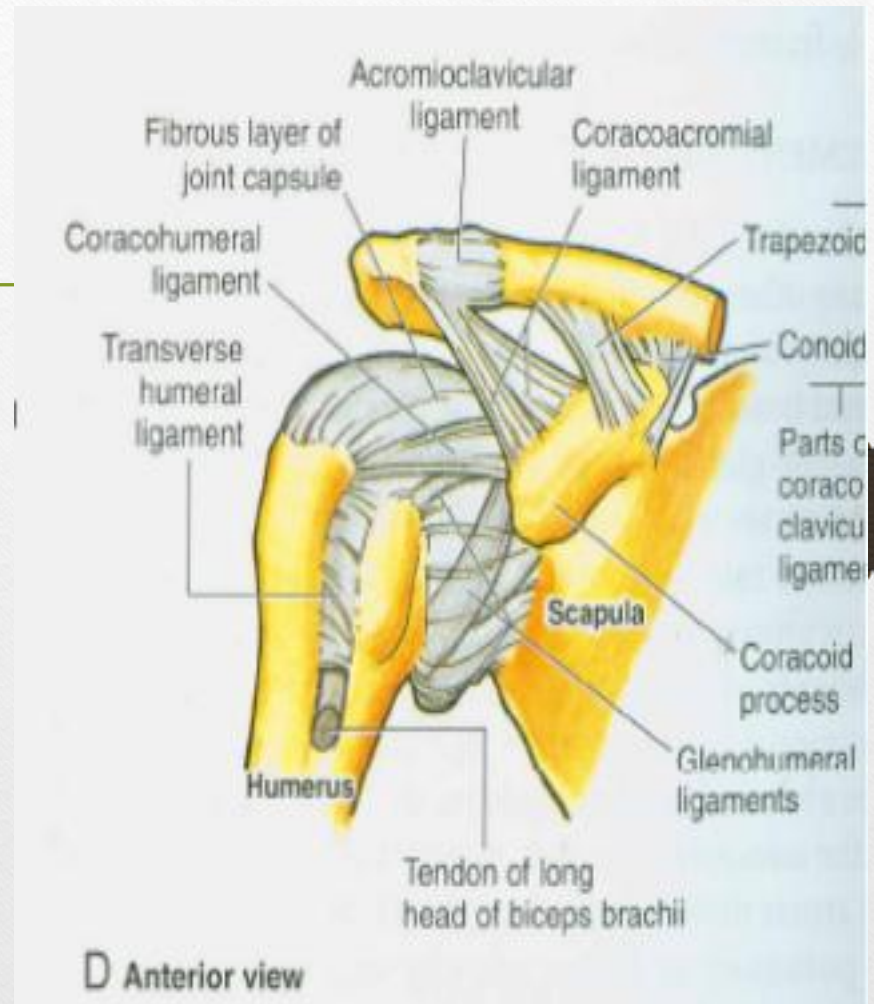


- Joint capsule

- attached to anatomical neck
- superiorly
- inferiorly
- synovial



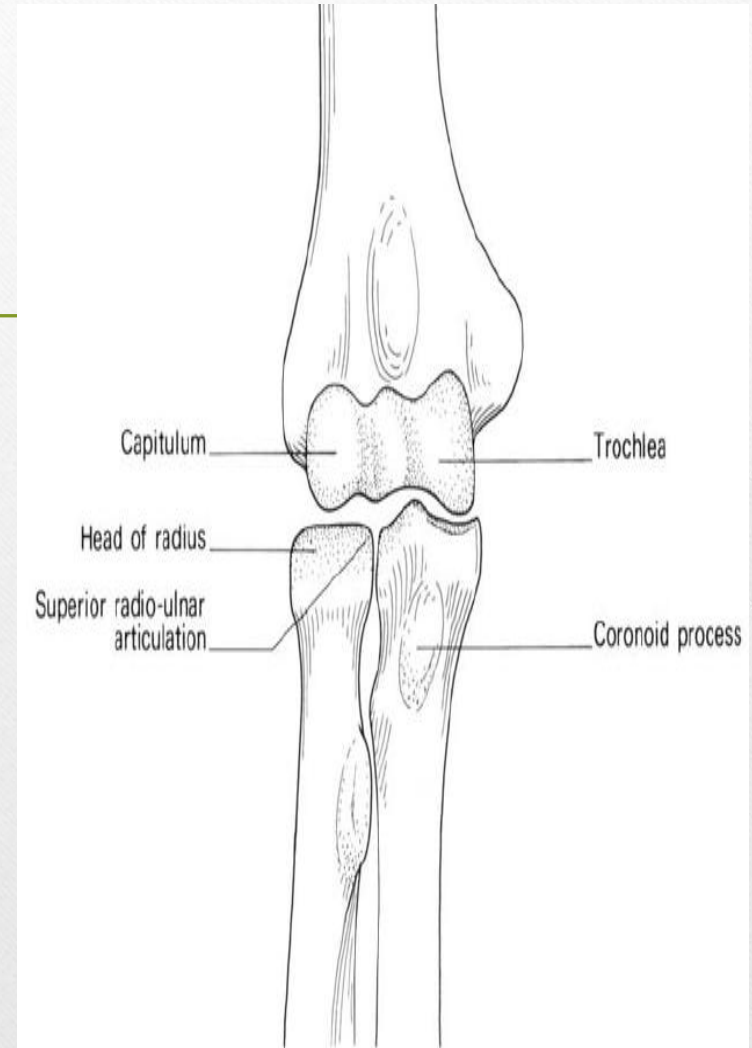
- Ligaments
 - **Glenohumeral ligament**
 - Strengthen anterior aspect of the capsule
 - **Coracohumeral ligament**
 - Strengthen superiorly
 - **Coracoacromial arch**
 - Formed by inferior aspect of acromion and coracoid process
 - Overlies the head of the humerus, prevents its superior displacement



- Movements: **flexion-extension**, **abduction-adduction**, **rotation** and **circumduction**
- **Blood supply**: anterior and posterior **circumflex humeral** arteries and branches of **suprascapular artery**
- **Innervation**: **suprascapular**, **axillary** and **lateral pectoral** nerves

Elbow joint

- Type: **Hinge synovial joint**
- Articulation
 - **Trochlea** and **capitulum** of humerus with trochlear **notch** of **ulna** and **head of radius**
- Joint capsule
 - Joint capsule surrounds the joint
 - Weak anteriorly and posteriorly but strengthened on each side by ligaments



- Ligaments

- **Radial collateral ligament**

- extends from the lateral epicondyle of the humerus and blends distally with annular ligament of the radius
-

- Encircles and holds the head of the radius in the radial notch of the ulna, forming the proximal radioulnar joint

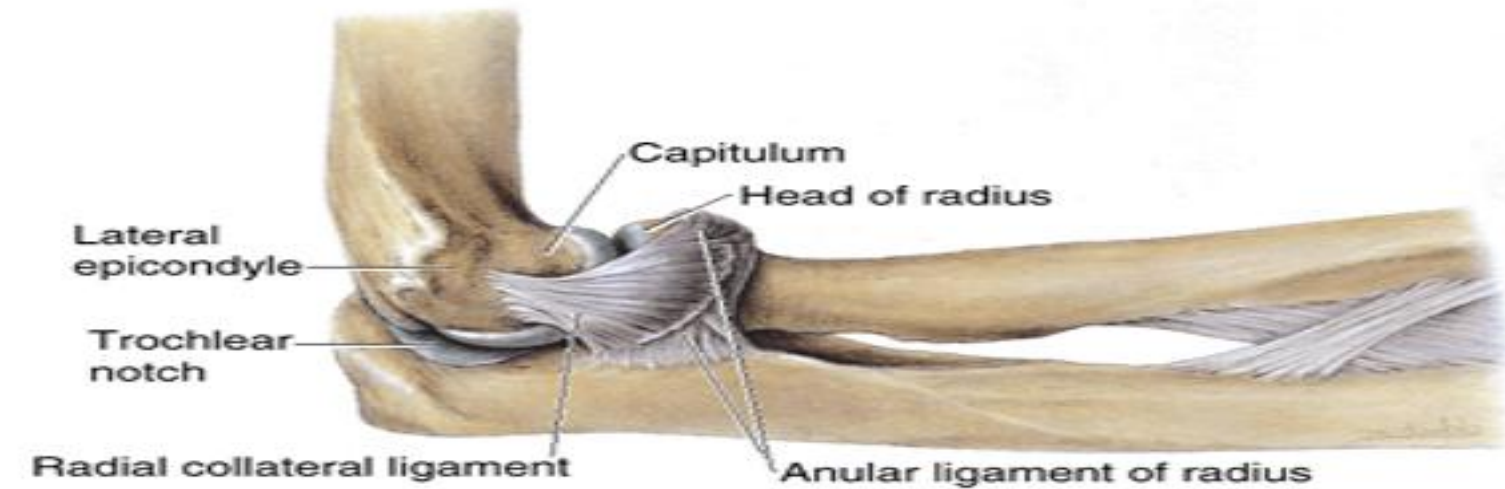
- **Ulnar collateral ligament**

- extends from the medial epicondyle of the humerus to the coronoid process and olecranon of the ulna

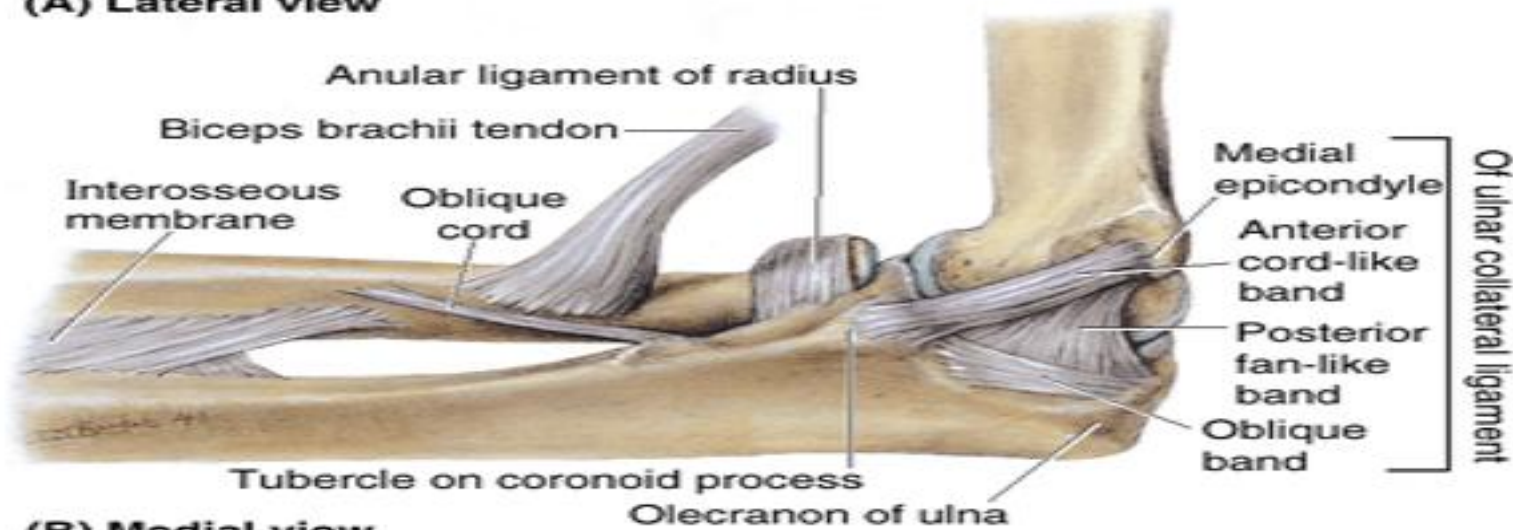
- Movements: Flexion and extension

- **Blood supply:** derived from the anastomosis of arteries around the elbow joint

- **Innervation:** musculocutaneous, radial, and ulnar nerves



(A) Lateral view

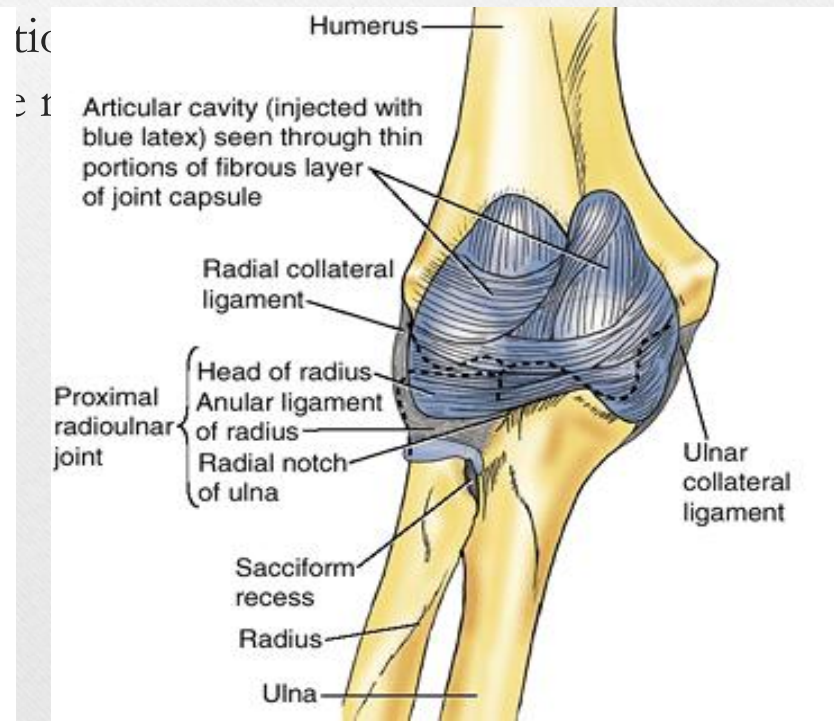
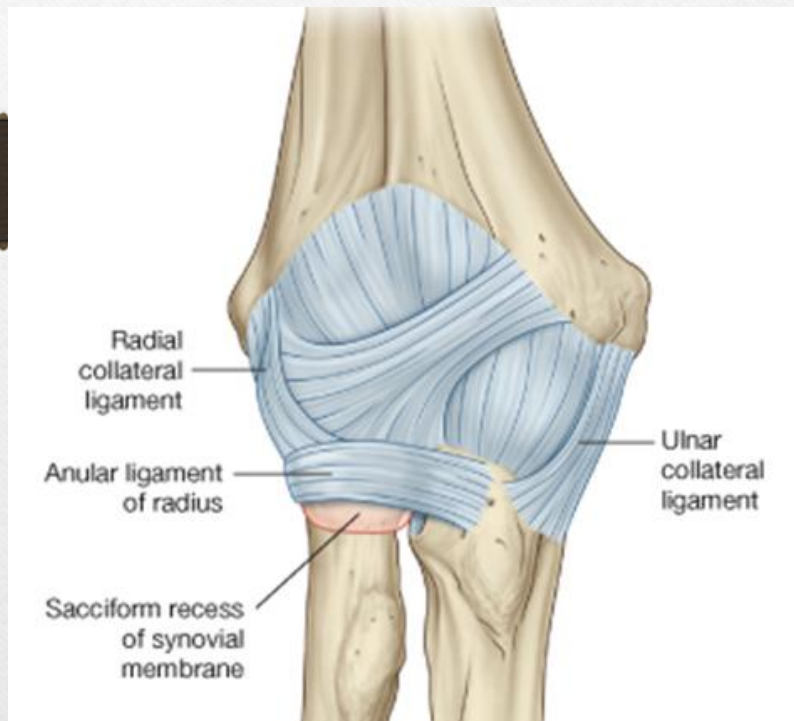


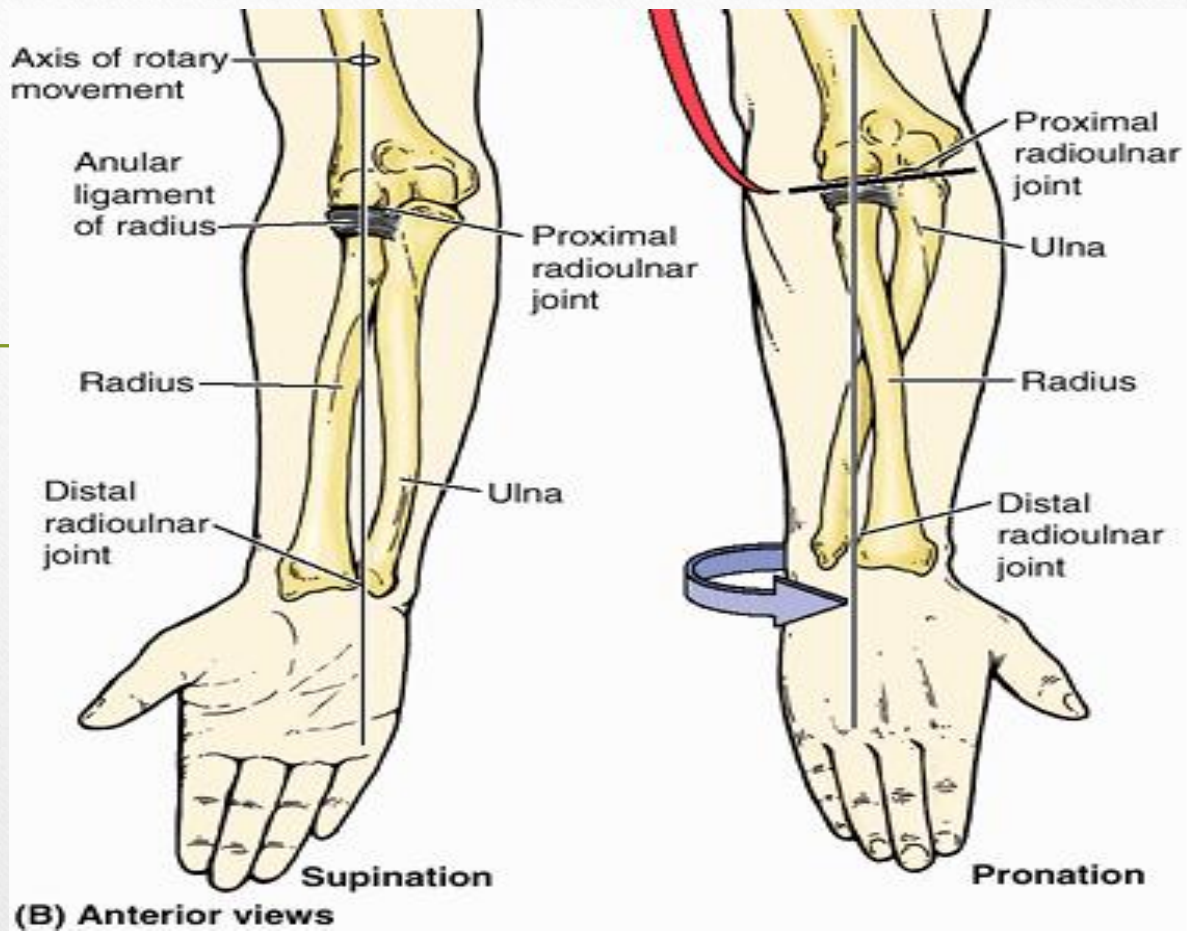
(B) Medial view

Proximal radio ulnar joint

- Type: **pivot synovial**
- **Articulation**
 - The head of **radius** articulates with radial notch of **ulna**
 - The radial head is held in place by the **annular ligament**
- **Joint capsule**
 - The fibrous layer of the joint capsule encloses the joint and is continuous with that of the elbow joint
 - The synovial membrane lines the internal surface of the fibrous layer and non-articulating aspects of the bones
- **Ligaments**
 - **Annular ligament**
 - attaches to the ulna, anterior and posterior to the radial notch, which forms a collar that, with the radial notch, forms a ring that completely encircles the head of the radius

- **Movements:** Pronation and supination
- **Blood supply:** the radial portion of the arterial anastomosis of the elbow joint
- **Innervation:** musculocutaneous, median, and radial nerves

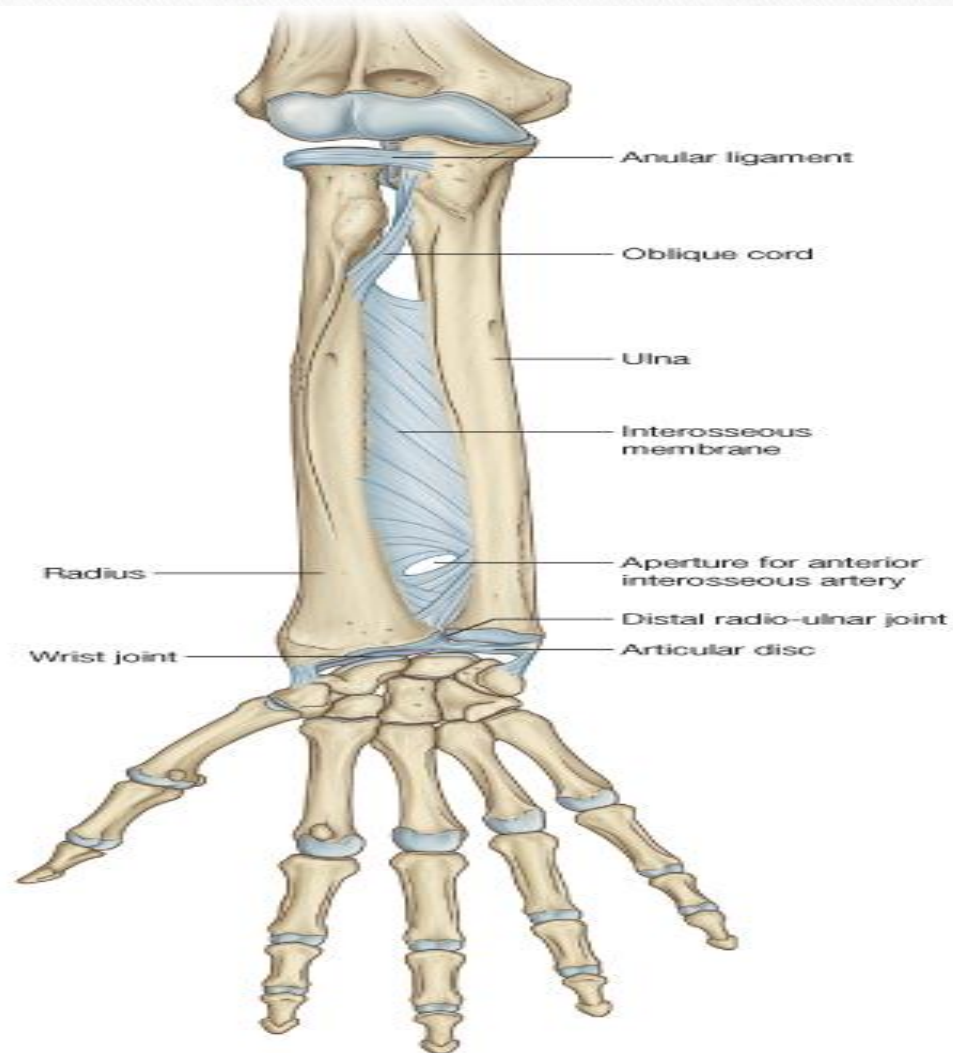




Movements of the Proximal Radioulnar Joint

Distal radio ulnar joint

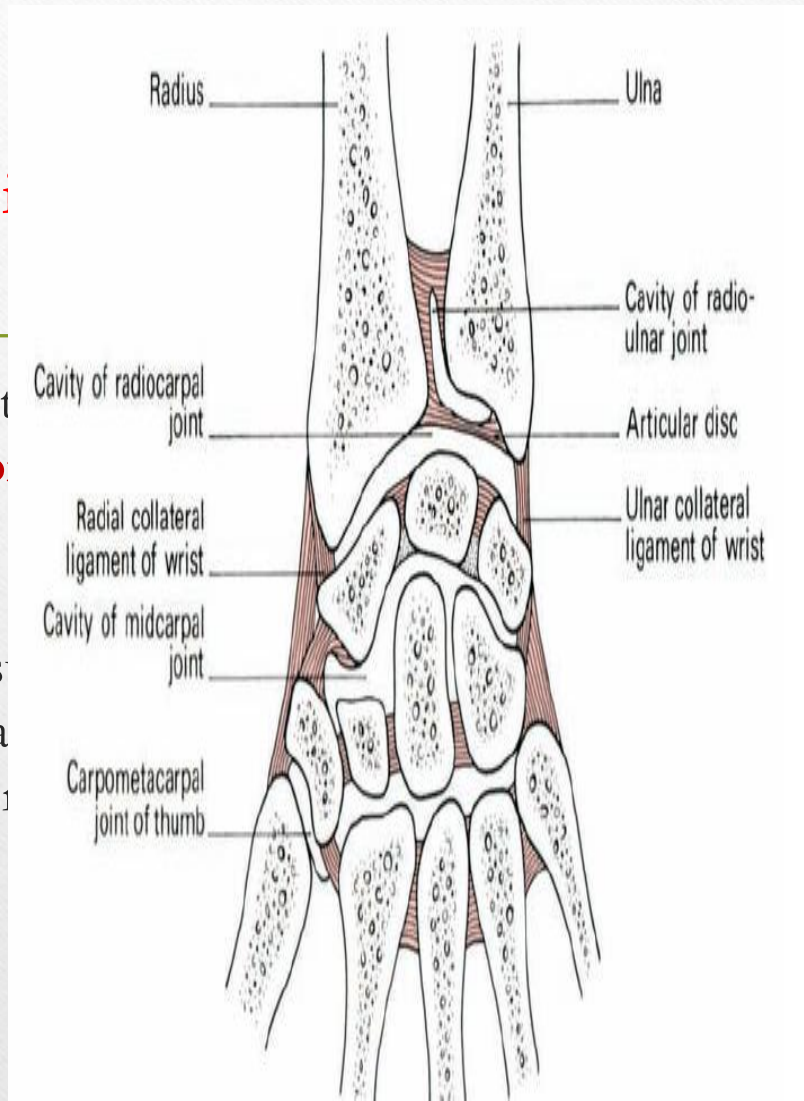
- Type: **Pivot synovial**
- **Articulation**
 - rounded **head of ulna** articulates with **ulnar notch** on the medial side of distal end of radius
 - fibrocartilaginous articular disc binds ulna and radius together
 - **articular disc separates the cavity of the joint from the cavity of the wrist joint**
- **Joint capsule**
 - The fibrous layer of the joint capsule encloses the joint but deficient superiorly
 - The synovial membrane extends superiorly between the radius and the ulna to form the recess



- Ligaments
 - **Anterior and posterior ligaments** strengthen the fibrous layer of the joint capsule
- Movements
 - The distal end of the radius rotates around the head of the ulna during **pronation** of the forearm and hand
- Blood supply: anterior and posterior **interosseous arteries**
- Innervation: anterior and posterior **interosseous nerves**

Wrist (radiocarpal) joint

- Type: **Condyloid synovial joint**
- Articulation
 - Distal end of **radius** and its art of carpal bones (**except pisiform**)
- Joint capsule
 - Fibrous layer of joint capsule s distal ends of radius and ulna a bones; lined by synovial memb

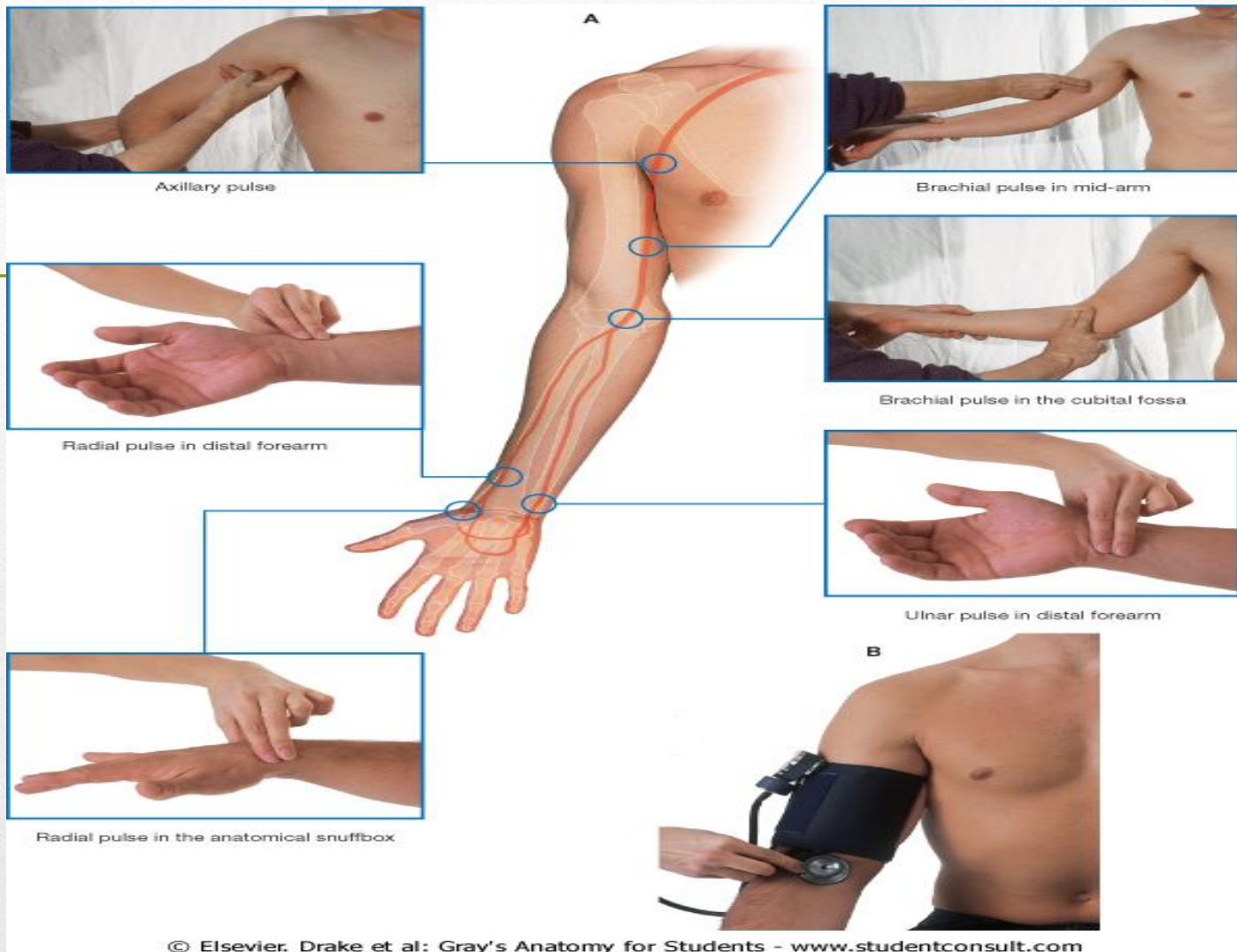


- Ligaments
 - **Anterior and posterior ligaments** strengthen fibrous capsule
 - **Ulnar collateral ligament** attaches to styloid process of ulna and triquetrum

- **Radial collateral ligament** attaches to styloid process of radius and scaphoid
- Movements: Flexion - extension, abduction - adduction, circumduction
- Blood supply: Dorsal and palmar carpal arches
- Innervation: Anterior interosseous branch of **median** nerve, posterior interosseous branch of **radial** nerve, and dorsal and deep branches of **ulnar** nerve

Pulse points in the upper limb

- **Axillary pulse:** axillary artery in the axilla
- **Brachial pulse** in mid-arm: brachial artery on the medial side of the arm. This is the position where a blood pressure cuff is placed
- **Brachial pulse in the cubital fossa:** brachial artery medial to the tendon of the biceps brachii muscle. This is the position where a stethoscope is placed to hear the pulse of the vessel when taking a blood pressure reading
- **Radial pulse** in the distal forearm: radial artery immediately lateral to the tendon of the flexor carpi radialis muscle. This is the most common site for 'taking a pulse
- **Ulnar pulse** in the distal forearm: ulnar artery immediately under the lateral margin of the flexor carpi ulnaris tendon and proximal to the pisiform.
- **Radial pulse** in the anatomical snuffbox: radial artery



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THANK YOU!!